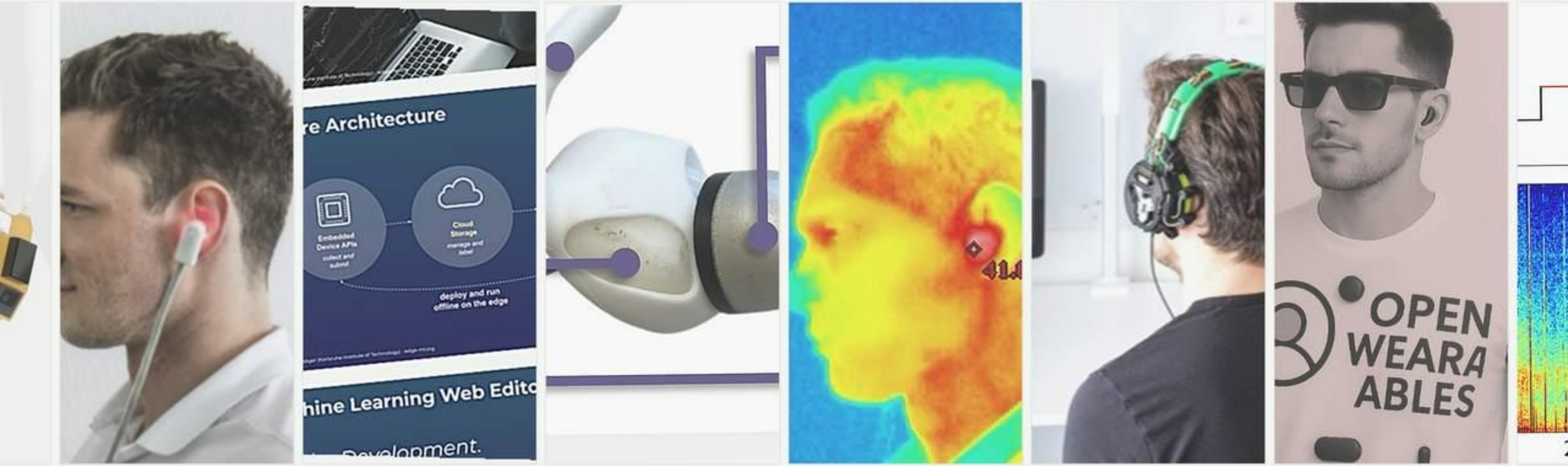


Human Computer Interaction SS26

Chapter
2
HIP

Michael Beigl, TECO

Annoucement:
OpenCast problems,
Zoom live stream (see ILIAS),
OpenCast Recordings (see ILIAS)

Lecture Overview (preliminiary)

Date / Day	Chapter	Topic
21 Apr 2026 (Tuesday)	Intro	
22 Apr 2026 (Wednesday)	HIP	Cognitive Foundations
28 Apr 2026 (Tuesday)	HIP	
29 Apr 2026 (Wednesday)	Exercise HIP / Lecture Integrated	
05 May 2026 (Tuesday)	Perception	
06 May 2026 (Wednesday)	Perception	
12 May 2026 (Tuesday)	Design Analysis	Design and Analysis
13 May 2026 (Wednesday)	Exercise Design Analysis and Perception	
19 May 2026 (Tuesday)	Design Analysis	
20 May 2026 (Wednesday)	Design	
26 May 2026 (Tuesday)	Pentecost	
27 May 2026 (Wednesday)	Pentecost	
02 Jun 2026 (Tuesday)	no lecture	
03 Jun 2026 (Wednesday)	Exercise Design	
09 Jun 2026 (Tuesday)	Design	
10 Jun 2026 (Wednesday)	Observation	Evaluation
16 Jun 2026 (Tuesday)	Observation	
17 Jun 2026 (Wednesday)	Exercise Observation	
23 Jun 2026 (Tuesday)	Interface and Interaction	Technology and Interaction
24 Jun 2026 (Wednesday)	Interfact and Interaction	
30 Jun 2026 (Tuesday)	Exercise Interface	

Processing (HIP) →

Chapter HIP Overview

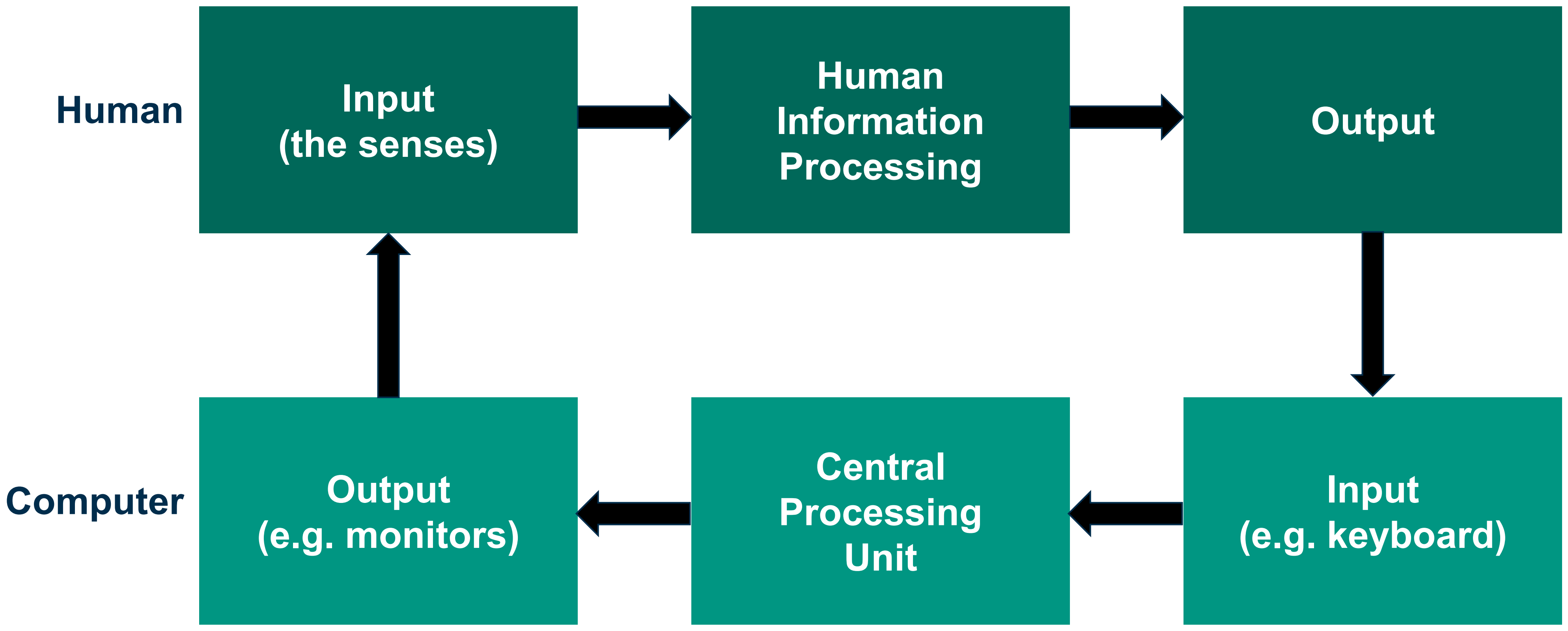
Human Information Processing

More info's:

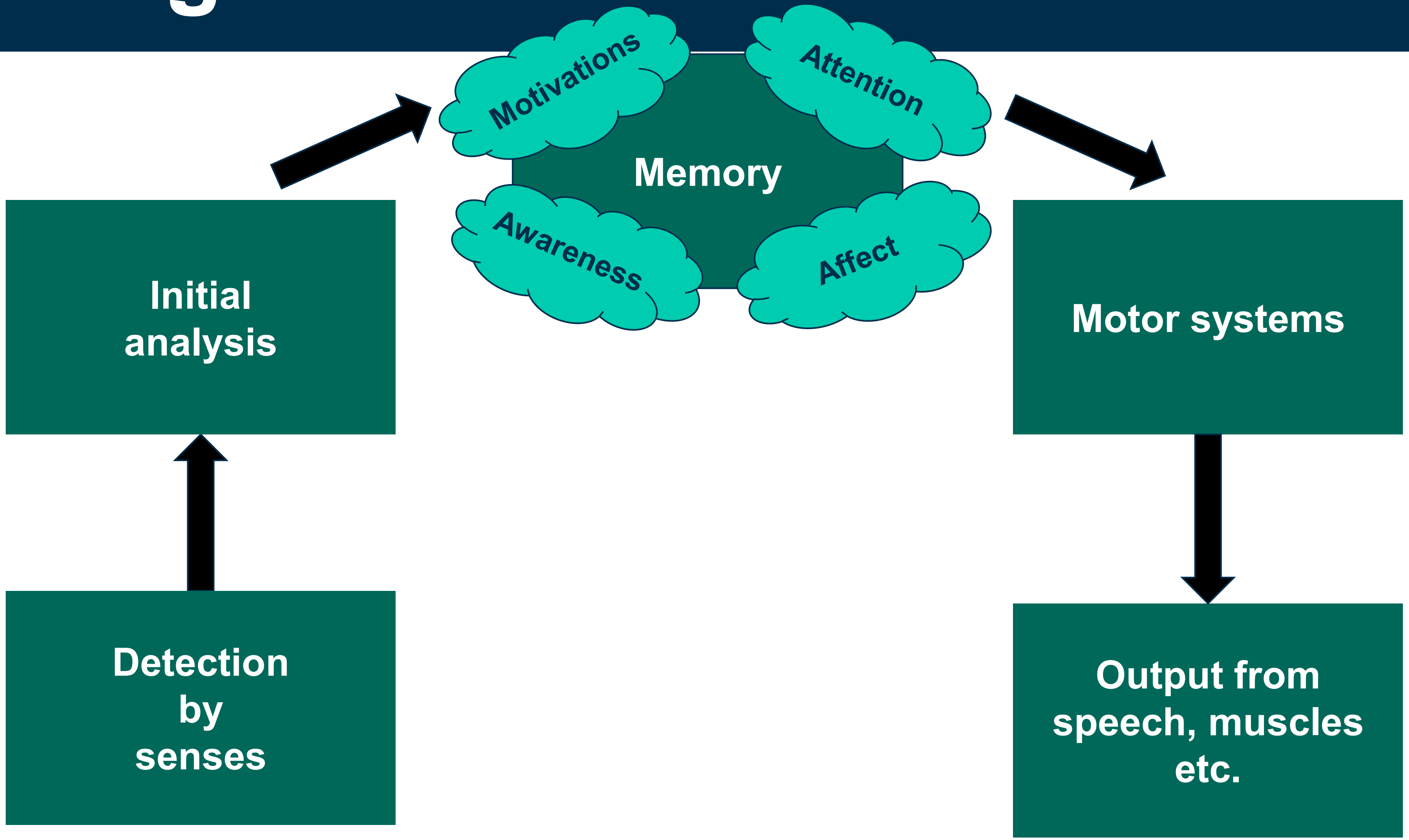
– Benyon 3rd edition chapter 21: Memory and attention
– Benyon 3rd edition chapter 23: Cognition and action
– Sanders chapter 3
– Newell, Card: The Prospects for Psychological Science in Human-Computer Interaction

HIP: Models

Human Information Processing: Old ACM Model

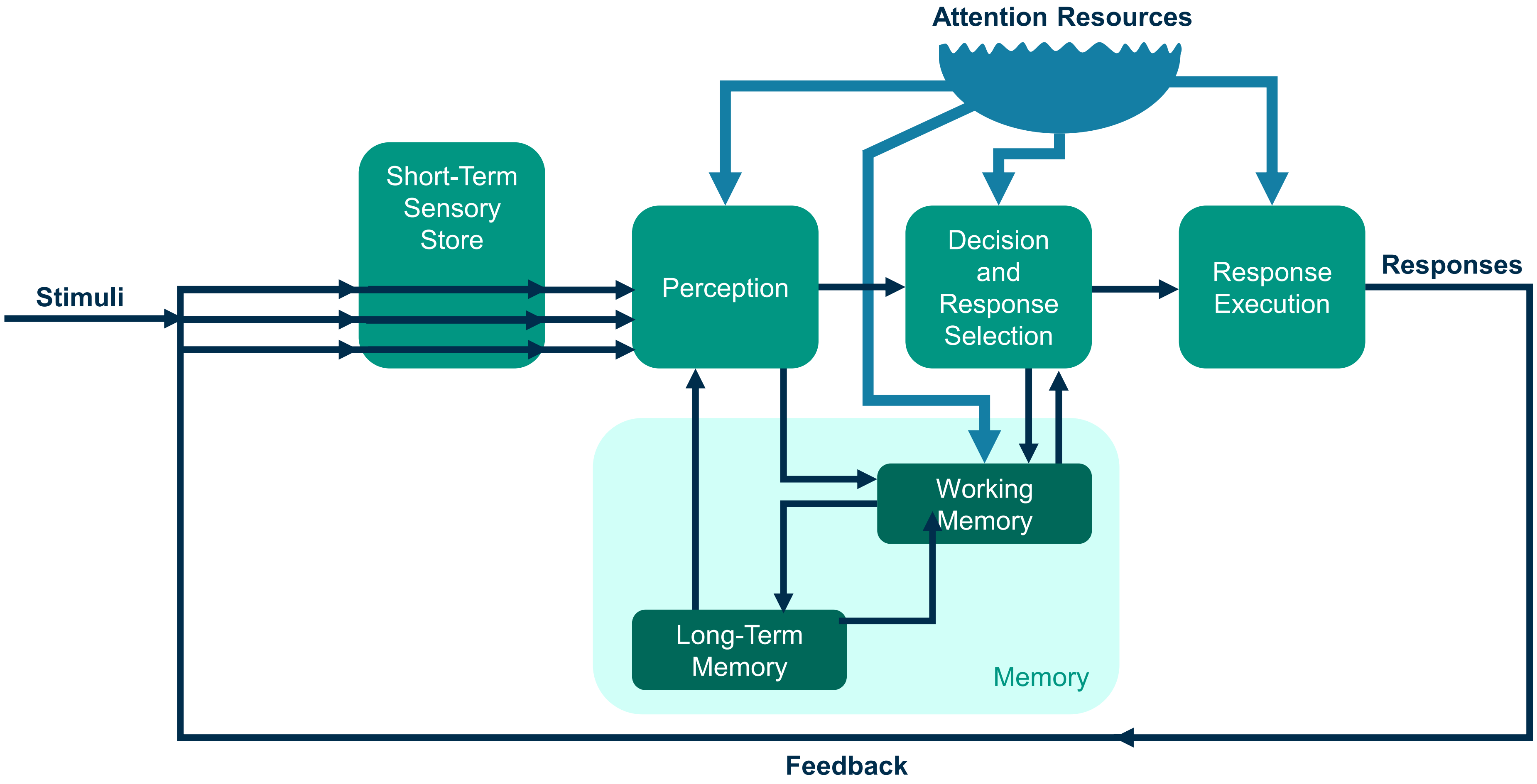


A simple Human Information Processing Model



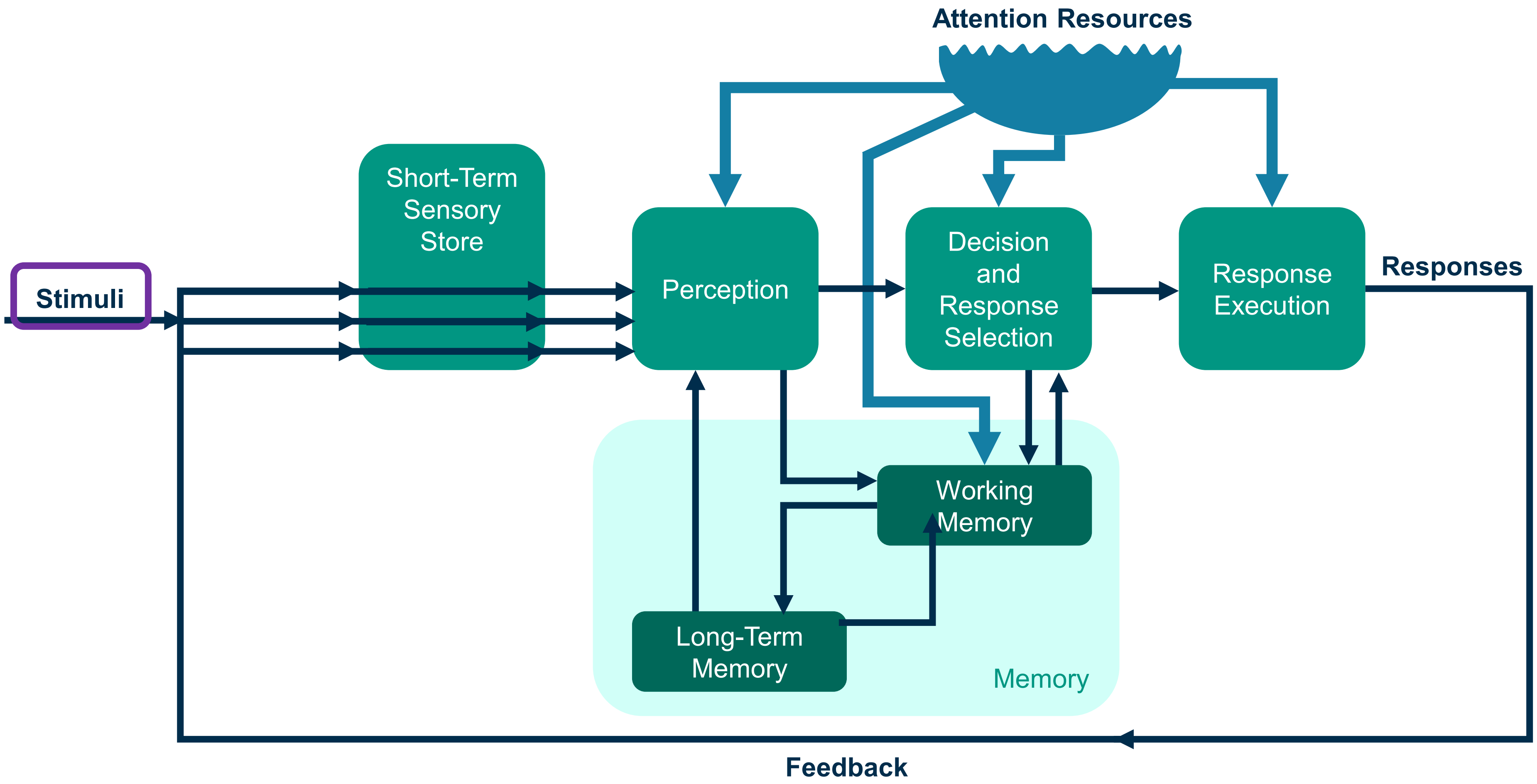
Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 101

Human Information Processing: Wickens and Hollands Model



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Human Information Processing



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Senses

Senses as our "windows to the world"

Definition: A sense is "a system that consists of a sensory cell type (or group of cell types) that respond to a specific kind of physical energy, and that correspond to a defined region (or group of regions) within the brain where the signals are received and interpreted."

What senses do you know?

Senses

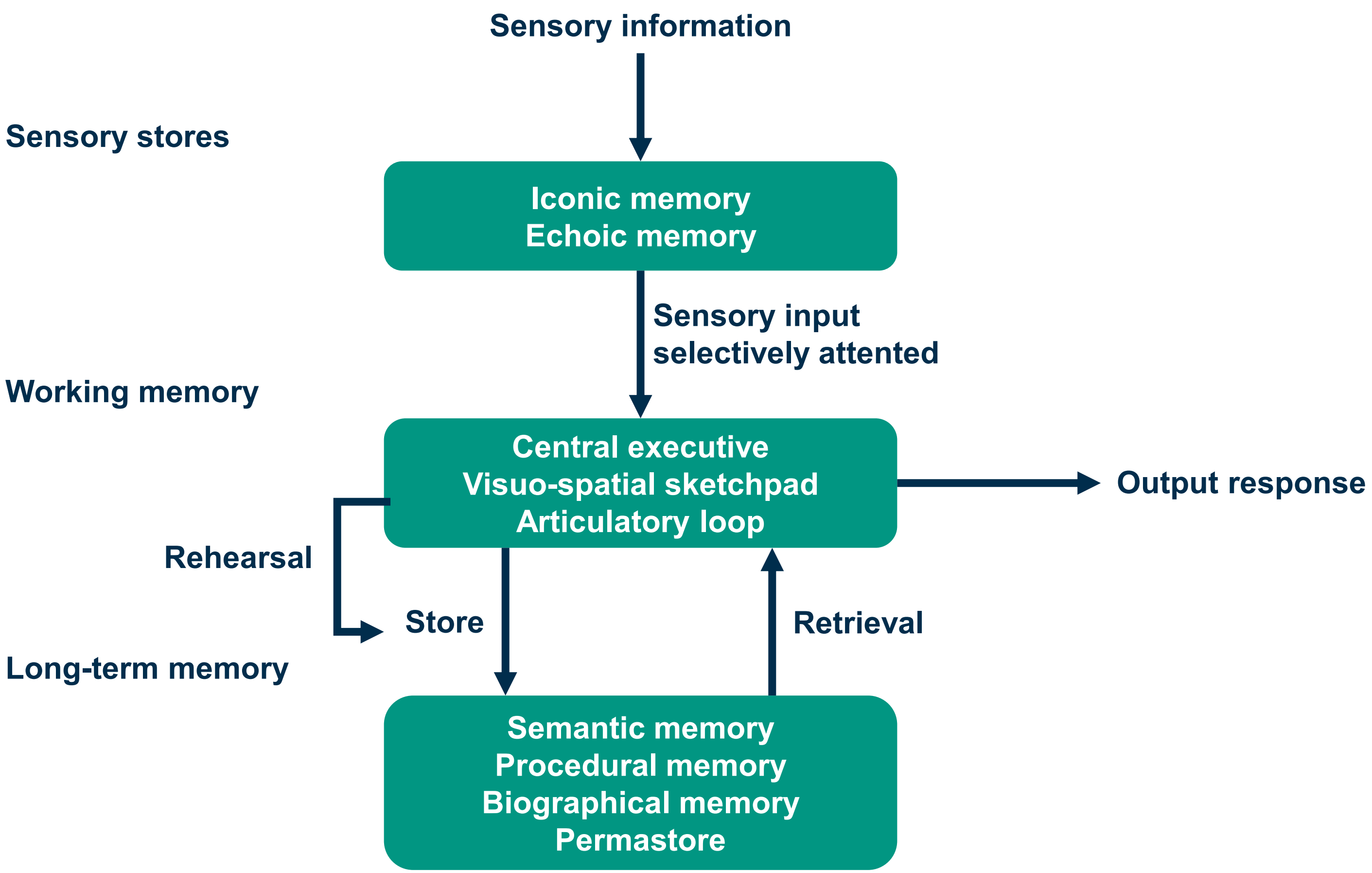
Senses as our “windows to the world”

Definition: A sense is “a system that consists of a sensory cell type (or group of cell types) that respond to a specific kind of physical energy, and that correspond to a defined region (or group of regions) within the brain where the signals are received and interpreted.”

What senses do you know?

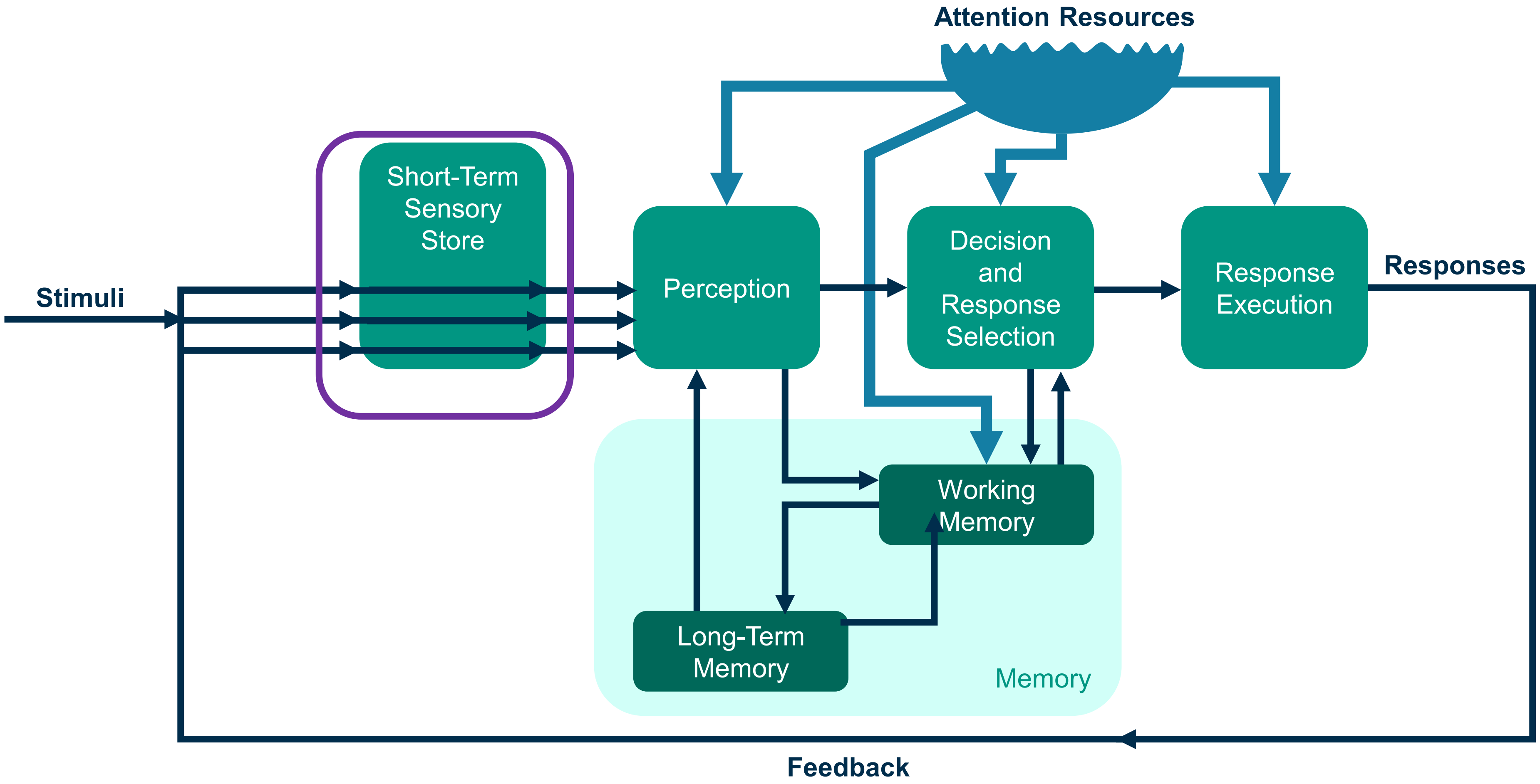
- six **external senses**: **sight, hearing**, touch, smell, taste, balance
- **internal senses**: thermoception, nociception (for pain), **proprioception** (Proprioceptive awareness of one's own body position)
- other beings have different and/or additional senses
- capabilities of our senses change over time (e.g. hearing)
- human senses are far from being perfect!
- senses limit what we can perceive – this can only, be extended by technical means!

Processing model: Learning: Senses, Memory



Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 101, graphically changed

Human Information Processing: STSS



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Short-Term Sensory Storage (STSS)

“Sensory Memory is the retention, for brief periods of time, of the effects of sensory stimulation.”
→ from Goldstein, p. 140

- collecting information for processing
- holding information briefly while initial processing is going on
- filling in the blanks when stimulation is intermittent

  → from Goldstein, p. 145

Cognitive Psychology: Connecting Mind, Research and Everyday Experience, E.B. Goldstein, 2004, Wadsworth Publishing, p. 136
online: http://64.78.63.75/samples/05PSY0304GoldsteinCogPsych.pdf

Short-Term Sensory Storage (STSS)

- sensory systems can “store” perceptions after the stimulus
- STSS does not need explicit attention, it works pre-attentive
- STSS is located in the brain
- STSS is not part of the conscious memory
- STSS != short term memory

two most important types of short-term sensory memory:

- **echoic memory** (lasts ~ **2-10 sec**. after stimulus)
- **iconic memory** (lasts ~ **0.5-1.0 sec**. after stimulus)

Examples:

- echoic memory: “What did you say?”
- iconic memory: grid of letters

Prepare for the Experiment

Test - Short-Term Sensory Storage (STSS)

Remember

DHFG
VJSA
LKOP
UHCW
POXQ

Test - Short-Term Sensory Storage (STSS)

Write down what you can remember from the letter rows!

Test - Short-Term Sensory Storage (STSS)

Who got everything right?

DHFG
VJSA
LKOP
UHCW
POXQ

Prepare for the Experiment

Test - Short-Term Sensory Storage (STSS)

Remember 3. Row

DHEG
VUWA
KHAR
MZBZ
CSQY

Test - Short-Term Sensory Storage (STSS)

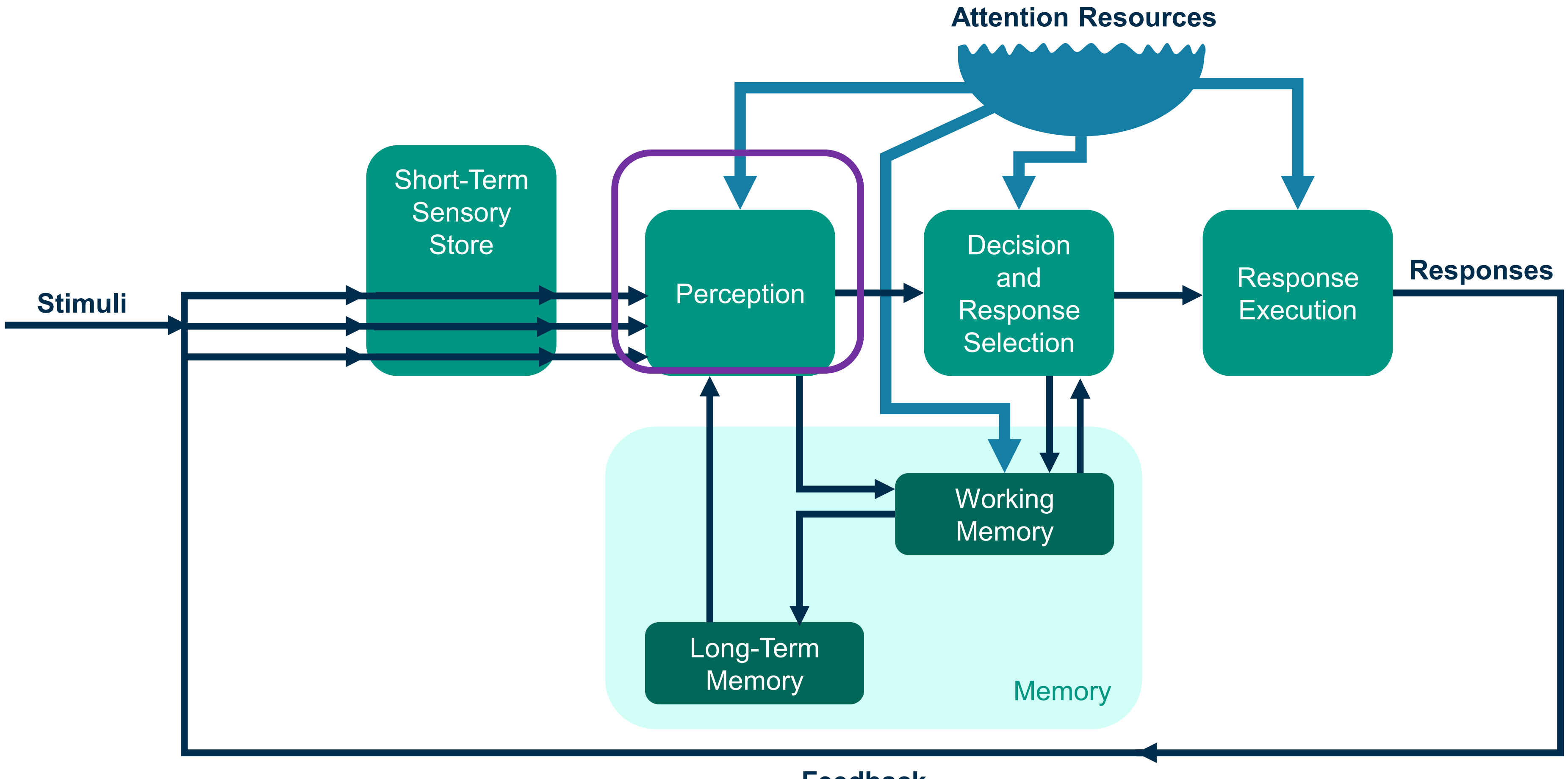
Write down what you can remember from the letter row!

Test - Short-Term Sensory Storage (STSS)

Who got the 3. row right?

DHEG
VUWA
KHAR
MZBZ
CSQY

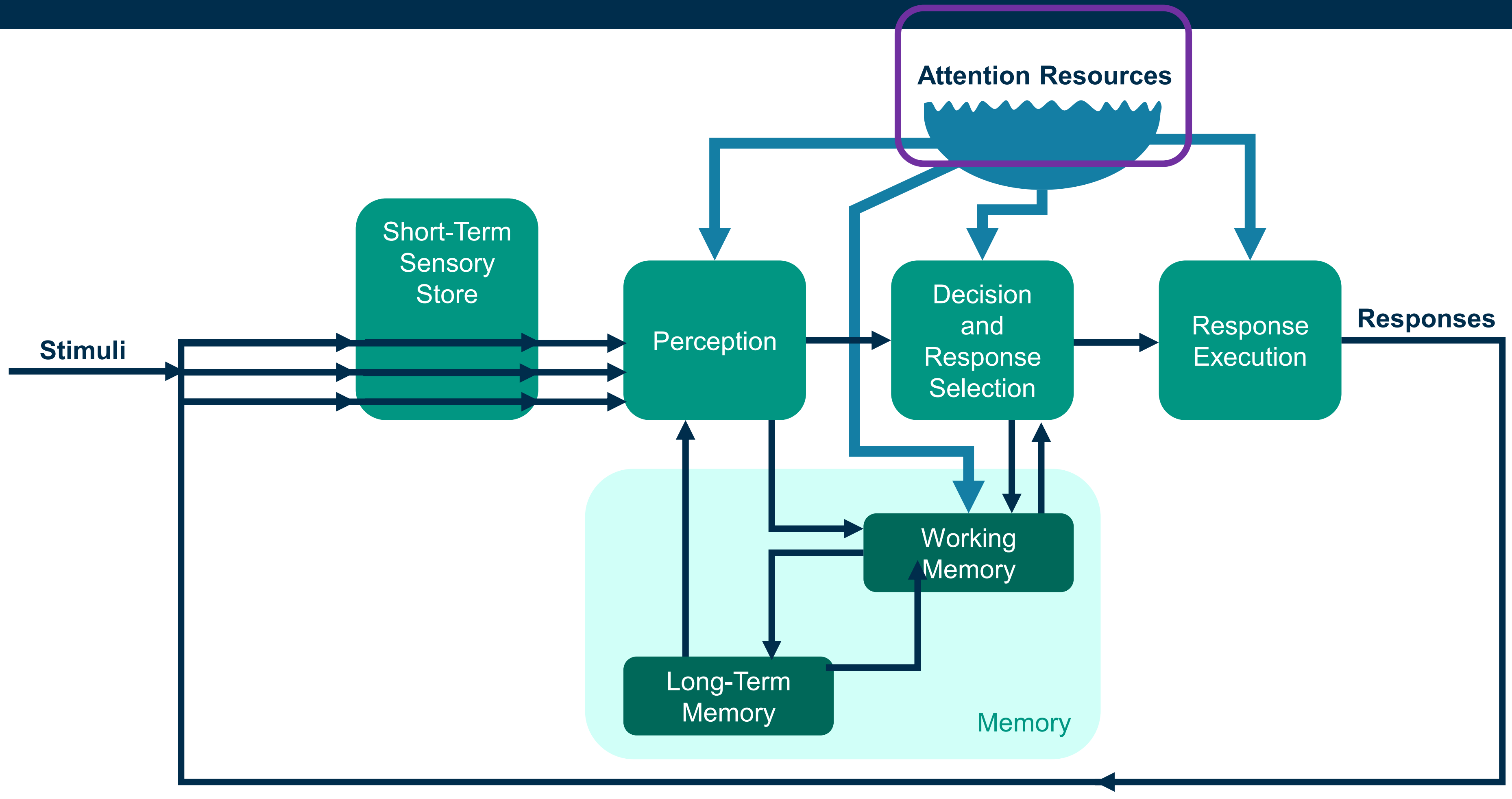
Perception



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

HIP: Attention

Attention



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Attention

Attention is the cognitive process of selectively concentrating on one aspect of the environment while ignoring other things.

attention as crucial element for cognitive processes as learning or task execution

types of attention:

- selective attention: willingly select focus
- focused attention: respond to external events
- divided attention: simultaneous focusing on different events

Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 12, 69-71

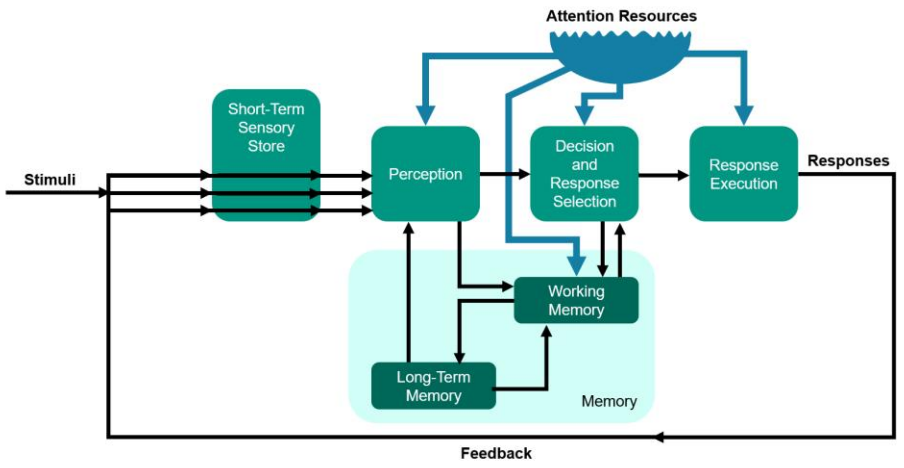
Attention Models

allocation model (Kahneman, 1973)

- limited amount of “processing power” at our disposal
- task execution depends on how much of our attention “capacity” we can spare on it

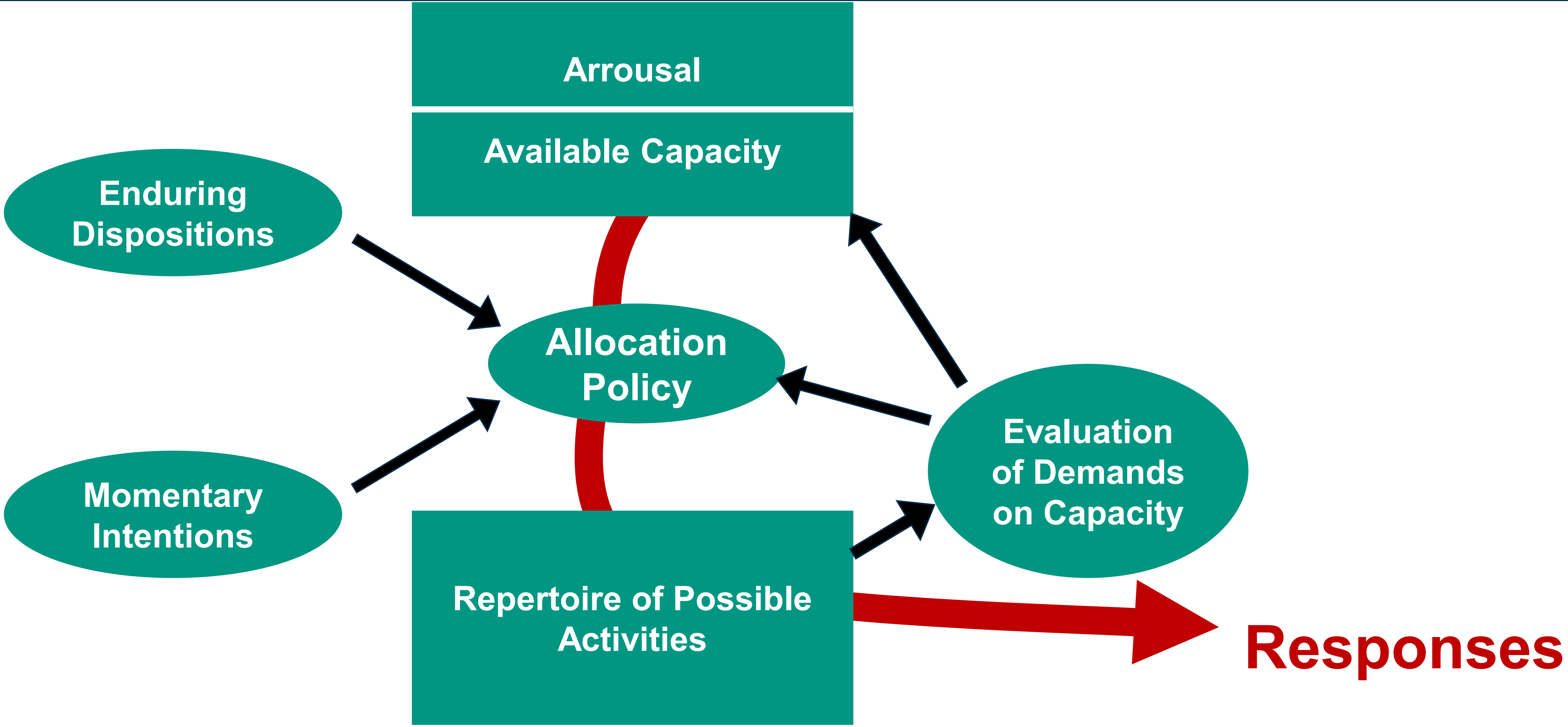
controlled and automatic attention (Schneider and Shiffirn, 1977)

- controlled processing makes heavy demands on attentional resources, is slow and limited in capacity, and involves consciously directing attention towards a task
- automatic processing makes no demands on attentional resources, is fast, unaffected by capacity limitations, unavoidable and difficult to modify, and is not subject to conscious awareness



Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 108-109

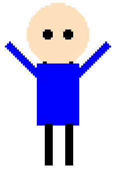
Capacity Model of Attention (Kahneman 1973)



Selective and Focused Attention

Concentration on one stimulus source

- needed for e.g. perception or learning (visual sampling or scanning)
- being too selective is referred to as “cognitive tunneling”
- negative examples: selecting cues that stand out rather than useful ones (e.g. during a presentation / argumentation)
- difficult: humans have tendency to be distracted
- important e.g. for directing attention to warning/error messages
- differences in errors between selective attention and focused attention: intentional selection of the wrong source vs. unintentional external influences



Divided Attention

- refer to the attention models
- limited attention capacity of humans
- important e.g. for layout of instruments in a cockpit (see Gestalt laws)
- differences in errors between focused attention and divided attention: some of our attention is directed to stimuli we do not wish to process vs. our limit to attend to all stimuli we wish to process
- → only a certain maximum of attention capacity can be divided to tasks

Time Experiment
A VOLUNTEER PLEASE!
Task: Name the colour of each line

Test - Divided Attention

Red

Green

Green

Red

Blue

Green

Red

Test - Divided Attention

Green

Red

Blue

Green

Red

Green

Red

Attention

- Attention can be directed at a particular task and/or divided between a number of different tasks
- Practice reduces the amount of attention required by a particular task. (e.g. typing blindly on a keyboard while reading a text)
- Attention and awareness are closely linked

Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 108

Pingo

Question:
Answer question about “Types of attention”

Link:
https://pingo.coactum.de/589809

HIP: Vigilance

Vigilance

"Vigilance is an aspect of attention which refers to detecting a rare event or signal in a desert of inactivity or noise"1

vigilance: detect signals over a long period of time; the signals are intermittent, unpredictable, and infrequent2

… but it is known that they happen

examples of vigilance tasks:

– security inspector x-raying luggage
– quality control in production

vigilance level (steady-state level performance)

vigilance decrement

[1]Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 109
[2]Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 34-36

Types of Vigilance

Vigilance types can be grouped in various ways

Typical Vigilance Paradigms Groups:

– Time: free-response, inspection (somehow interval)

– Order: simultaneous, subsequent

– Perception: Sensory, Cognitive

Vigilance Paradigms Group 1: time

free-response paradigm:

- a target event occurs at any time
- non-events are not defined
- example: **power plant monitor supervision**

inspection paradigm:

- events occur at fairly regular intervals
- some are events, most are non-targets
- example: **quality control (most items are ok, only some have defects)**

Vigilance Paradigms Group 2: order

successive vigilance paradigm:

- target stimulus has to be remembered
- successive events have to be compared to the target stimulus
- example: **detect if a color is darker than the initial target**

simultaneous vigilance paradigm:

- all events/information needed for discrimination are present at the same time
- example: **compare many types of garment to a standard piece of fabric**

Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 33-36

Vigilance Paradigms Group 3: perception

sensory vigilance paradigm:

- signals represent changes in the auditory or visual intensity
- example: **color changes**

cognitive vigilance paradigm:

- signals represent “information”: symbolic or alphanumeric stimuli
- example: **proofreading a manuscript**

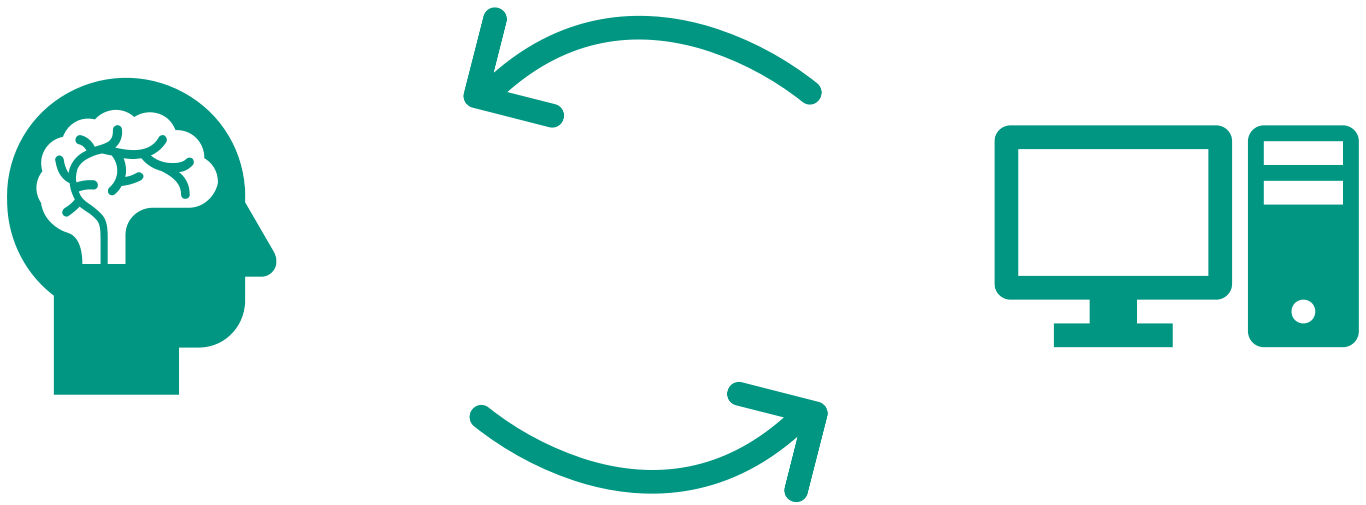
Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 35

Workload and Measurement: Signal Decision Theory

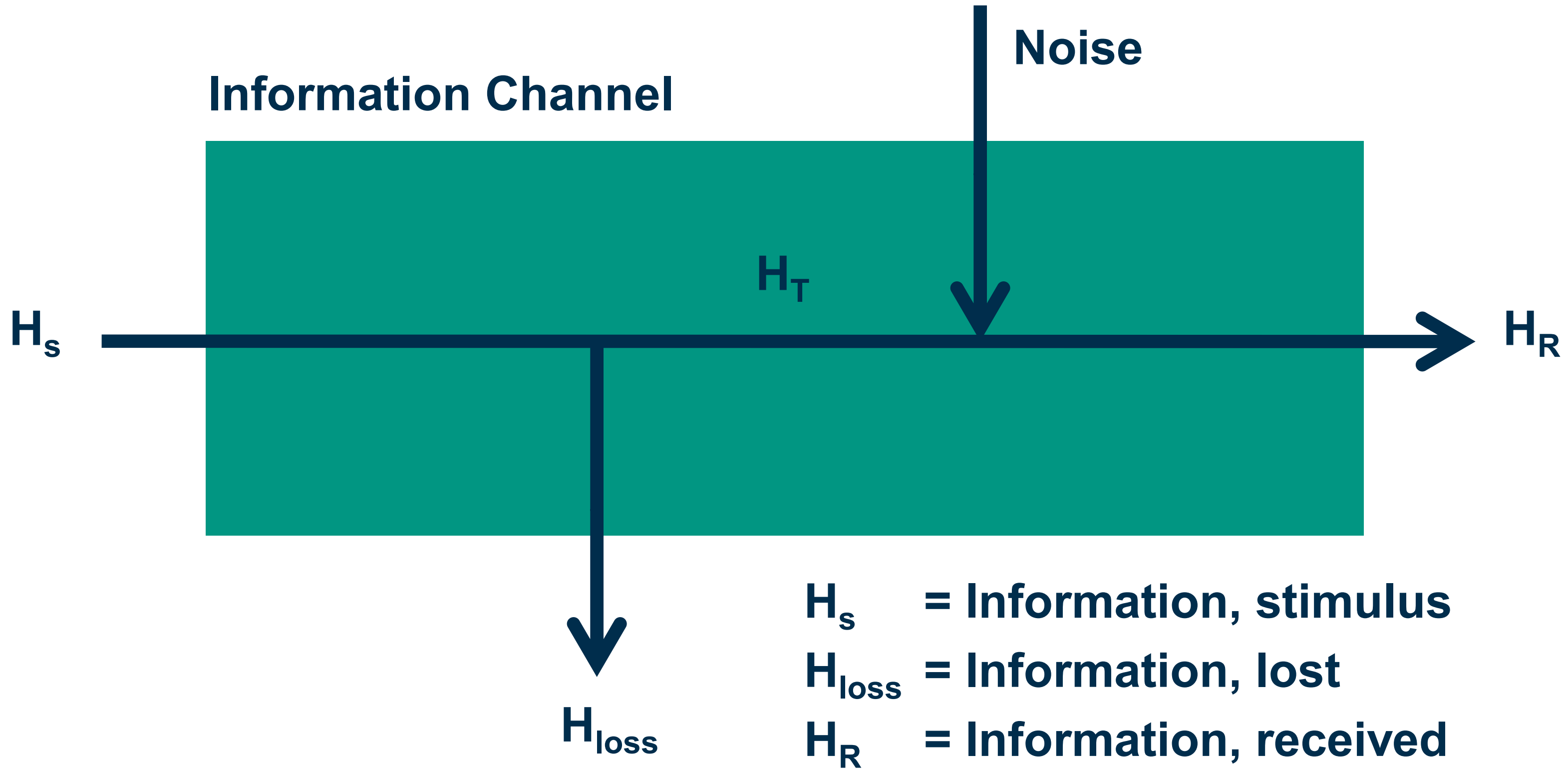
Information Transmission

Human capability of information processing is limited!

- **it is not only of interest, how much information is presented to a human, but:**
- how much information is transmitted from stimulus to response
- capacity of the information channel
- how rapidly information is transmitted (bandwidth)



Information Channels



H_s = Information, stimulus
H_{loss} = Information, lost
H_R = Information, received
H_T = Information, transmitted

Motivation

- Human-Computer Interaction involves information processing on the human side
- An event (signal) occurs in the environment; the human perceives it, processes it and reflects upon it, and transmits this information to a response.
- Signals are stimuli that are perceived by our senses and transformed in neural activity in the brain
- Info-Theory: Able to describe single process in brain, but also complex human reaction

Event occurs
e.g. Fire

Perceive it
e.g. sees fire

Processes it
e.g. "Oh there is a fire"

Reflects it
e.g. "I need to extinguish the fire"

Transmit to a response
e.g. extinguish the fire

Signal Detection Theory
Confusion Matrix

- two discrete states: signal or noise
- two possible responses: yes or no

4 possible classifications

state of the world

response	Signal	Noise
Yes	Hit	False alarm
No	Miss	Correct rejection

Signal Detection Theory
Confusion Matrix

- two discrete states: signal or noise
- two possible responses: yes or no

4 possible classifications

Confusion Matrix:

response \ state of the world	Signal	Noise
Yes	Hit	False alarm
No	Miss	Correct rejection

Signal Detection Theory → Classification Test in ML

- two discrete states: signal or noise
- two possible responses: yes or no

→ 4 possible classifications

- confidence in classification are expressed as probabilities:

Confusion Matrix:

response \ state of the world	Signal	Noise
Yes	TruePositive (TP)	FalsePositive (FP)
No	FalseNegative (FN)	TrueNegative (TN)

Precision: TP/(TP+FP)

Recall= Sensitivity= TP/(TP+FN)

Selectivity= Specificity= TN/(TN+FP)

F1= harmonic mean of Precision and Recall

Accuracy = (TP+TN) / all

Signal Detection Theory

The SDT model assumes that there are two stages of information processing in the task of detection:

- sensory evidence is aggregated concerning the presence or absence of the signal
- a decision is made about whether this evidence indicates a signal or not

Signal detection & neural activity

- external stimuli generate neural activity
- on average, there will be more neural evidence when a signal is present than when it is absent

Signal Detection Theory

Types of noise are:

- external (in the data)
  → e.g. a picture that you watch contains not only the information to be detected
- internal (in our brain, e.g. firing rate of neurons varies for the same stimulus)

Recognition process leads to

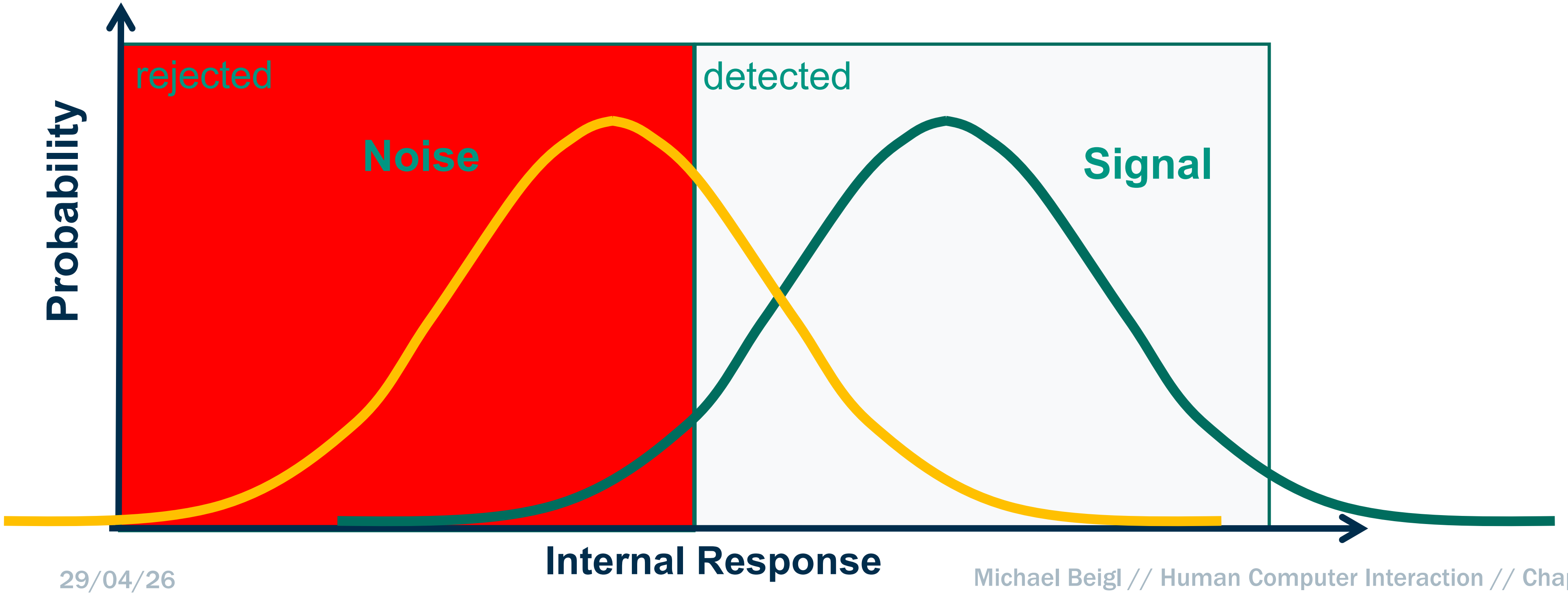
- internal response (in the observer's brain)
  → which is noisy as well → noise

E.g. snow overlaying a sign

▶ image source

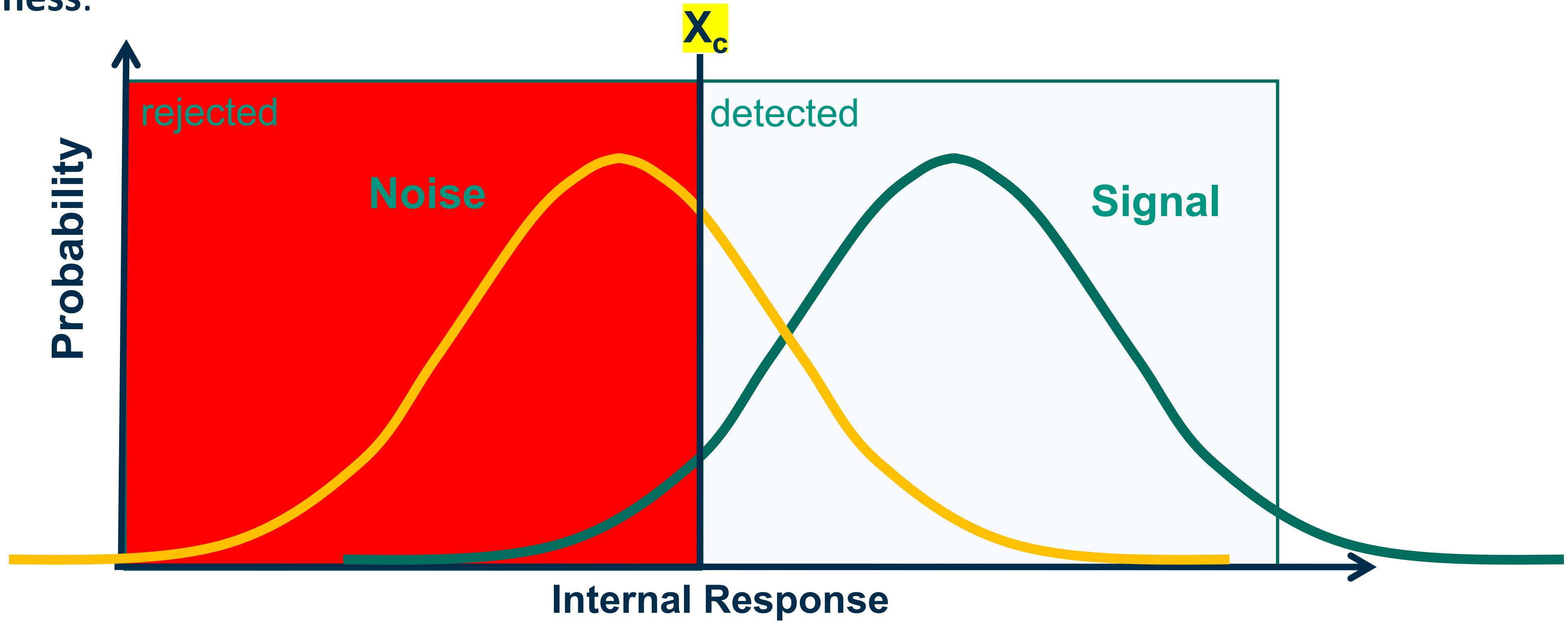
Signal Detection Theory

- The probability distributions for internal response can be different for noise and for signal!
- Internal response as neural activity resulting from a stimulus



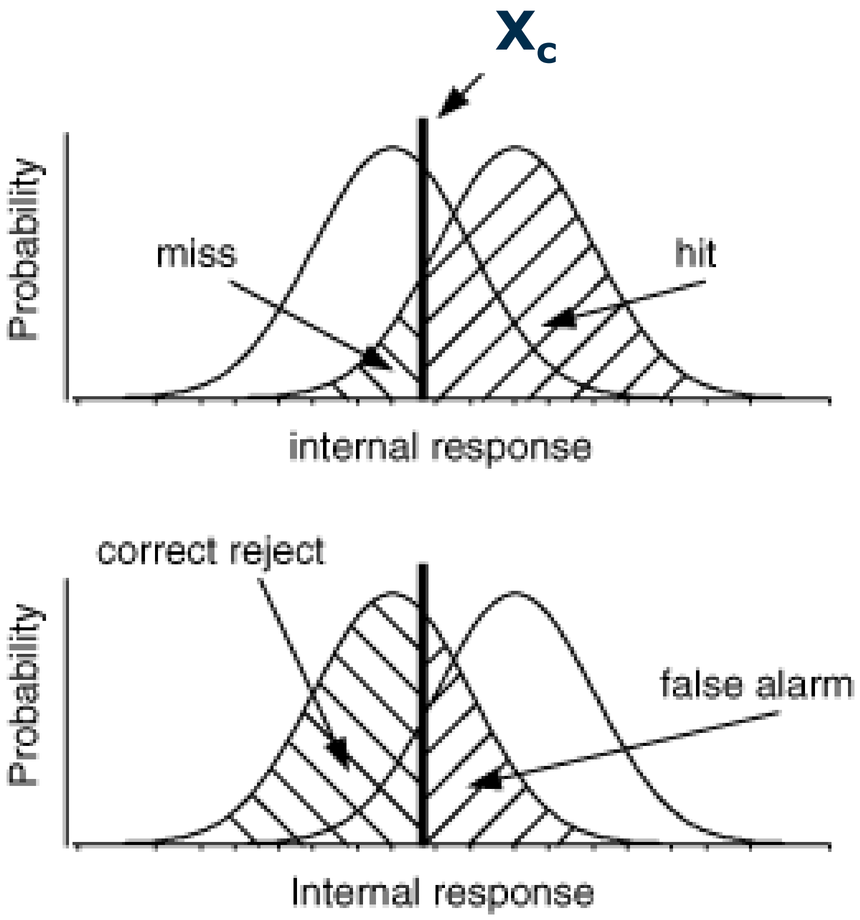
Signal Detection Theory sluggish beta

- Humans tend to set X_c not optimal
- **sluggish beta** refers to a system or interface parameter that **adapts too slowly** **to changes in user behavior or context or that is** **set wrong** **due to external signals, resulting in delayed or suboptimal responsiveness**.



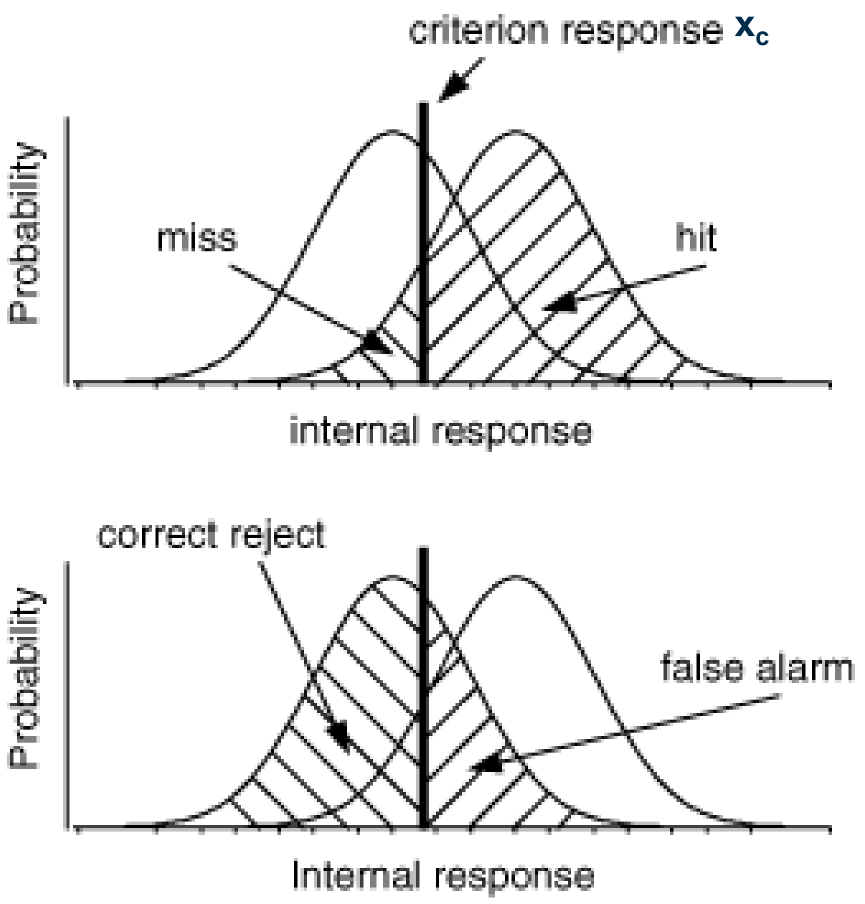
Signal Detection Theory

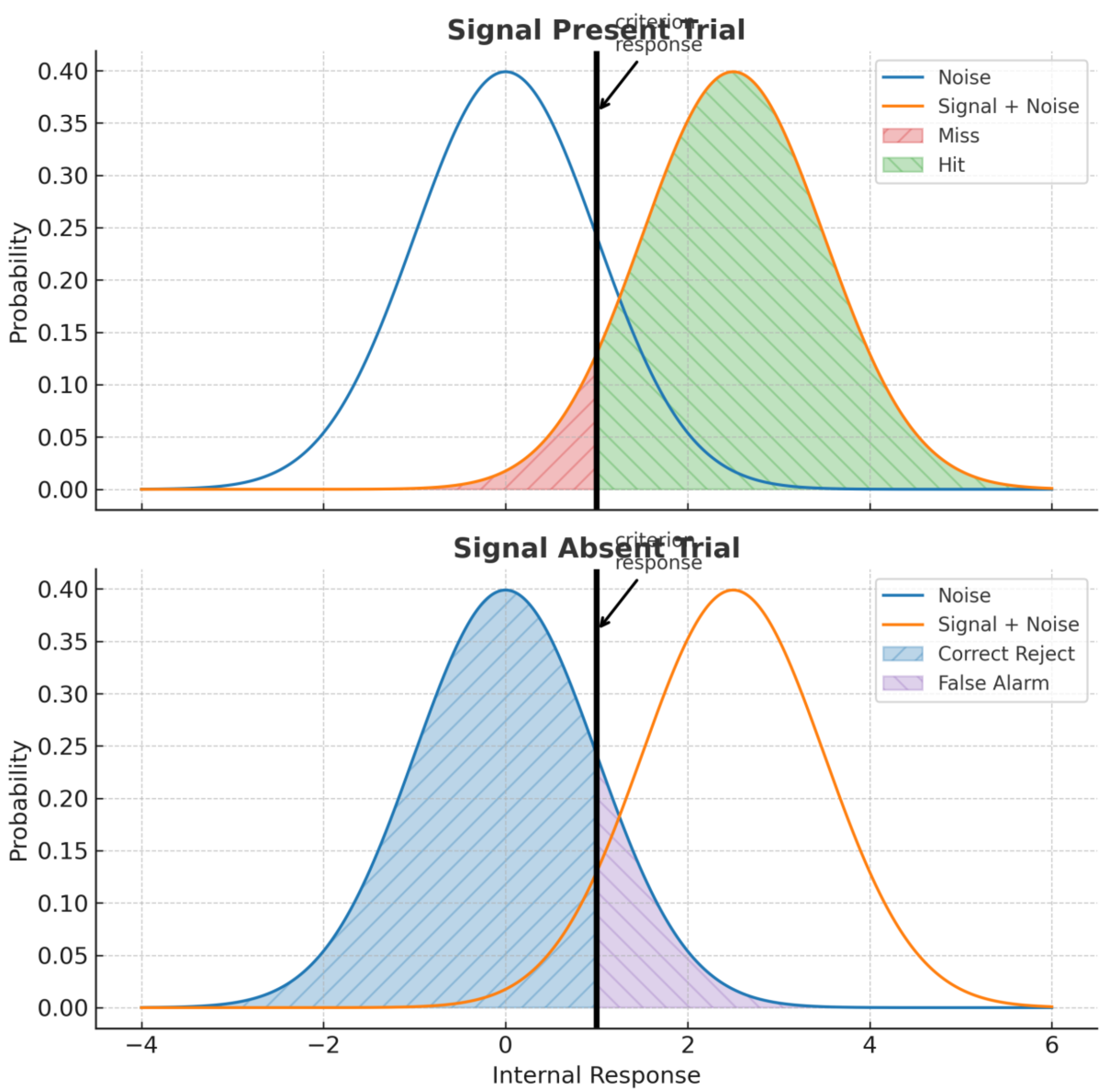
signal classifications revisited:
- Hit
- Miss
- False alarm
- Correct rejection



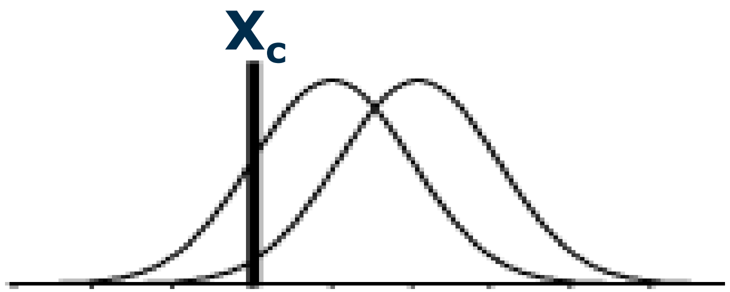
Processing (HIP) →

Which representation is better? And why?

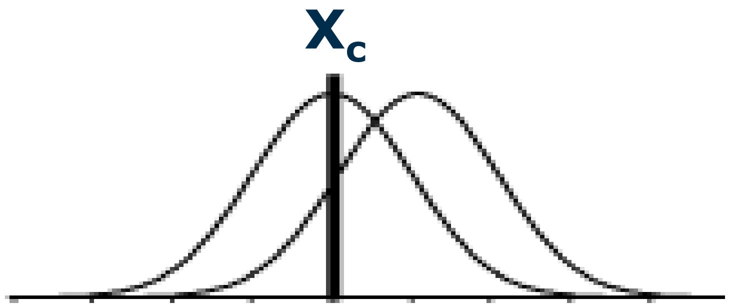




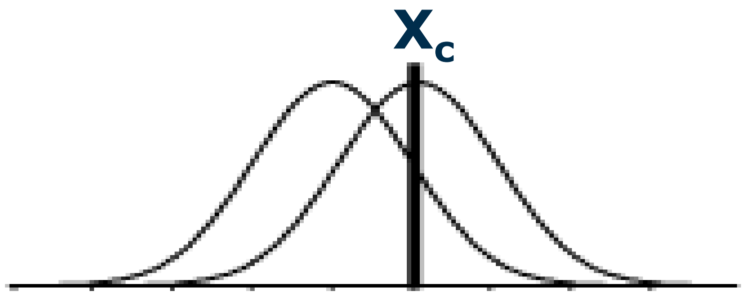
Signal Detection Theory
Examples



Hits = 97.5% High Recall (condition:P(S)=0.5))
False alarms = 85% Low Selectivity (0.15)



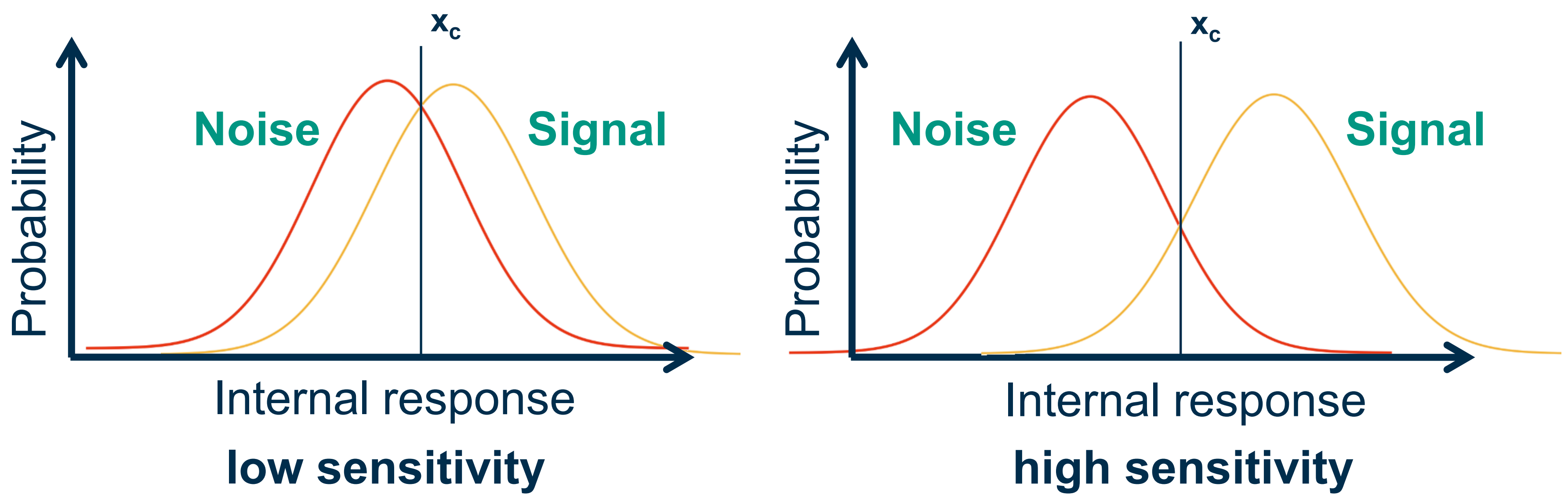
Hits = 84%
False alarms = 50%



Hits = 50% High Selectivity (0.84)
False alarms = 16% Low Recall (0.5)

Signal Detection Theory

Low vs. High Sensitivity of Signals



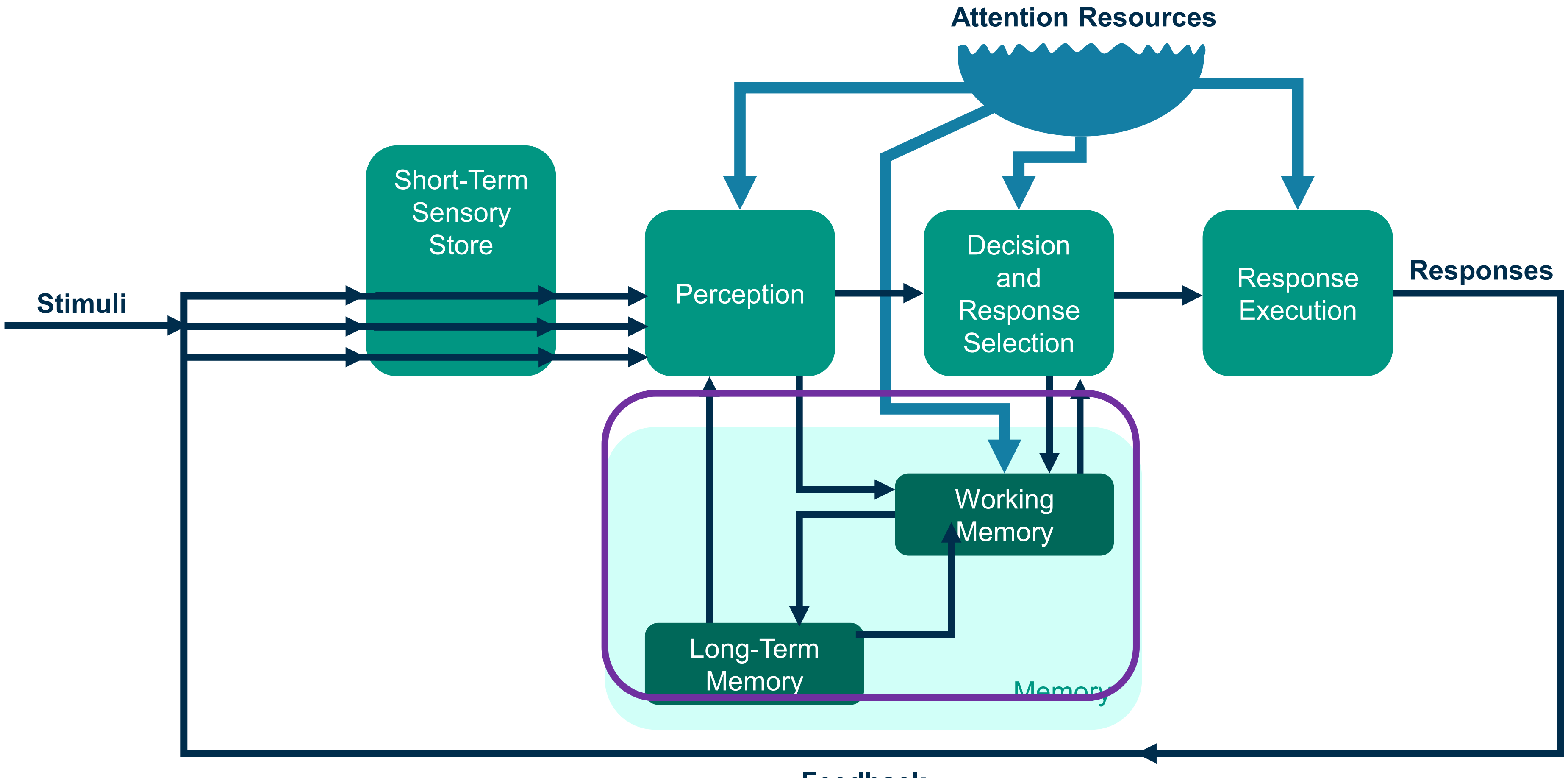
Pingo

Question:
Answer question about “Signal Detection Confusion Matrix”

Link:
https://pingo.coactum.de/589809

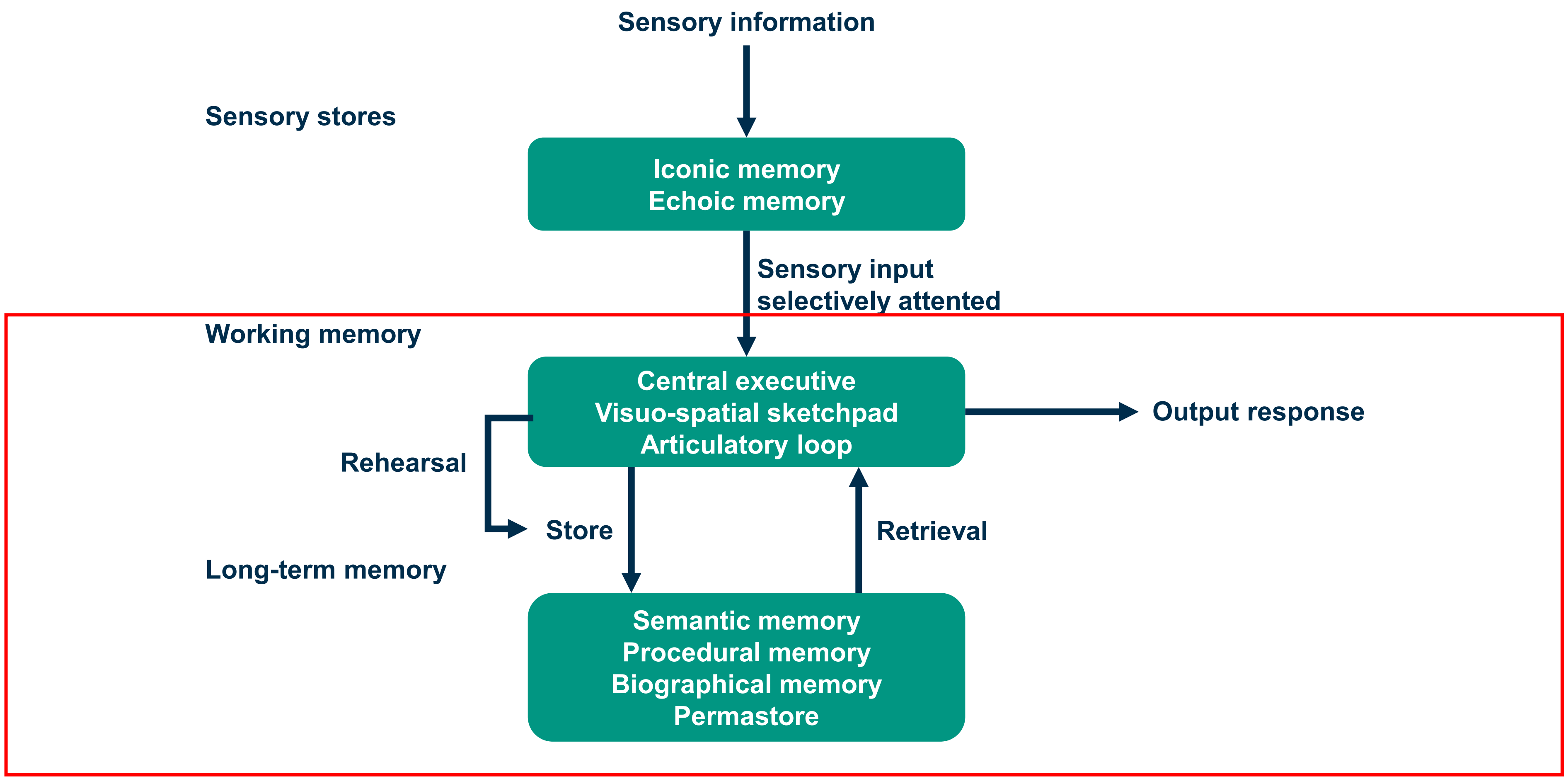
Memory →

Memory



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Memory and Senses: Encoding and Retrieval of Information



Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 101, graphically changed

Memory

human memory consists of two major components:

- working memory (formerly: short-term memory)
- long-term memory

memory “processes”:

- Recall
- Recognition
- Chunking
- Rehearsal
- Forgetting
- NOT: database (Nuremberg Funnel)

Human memory is multi-modal!

Memory

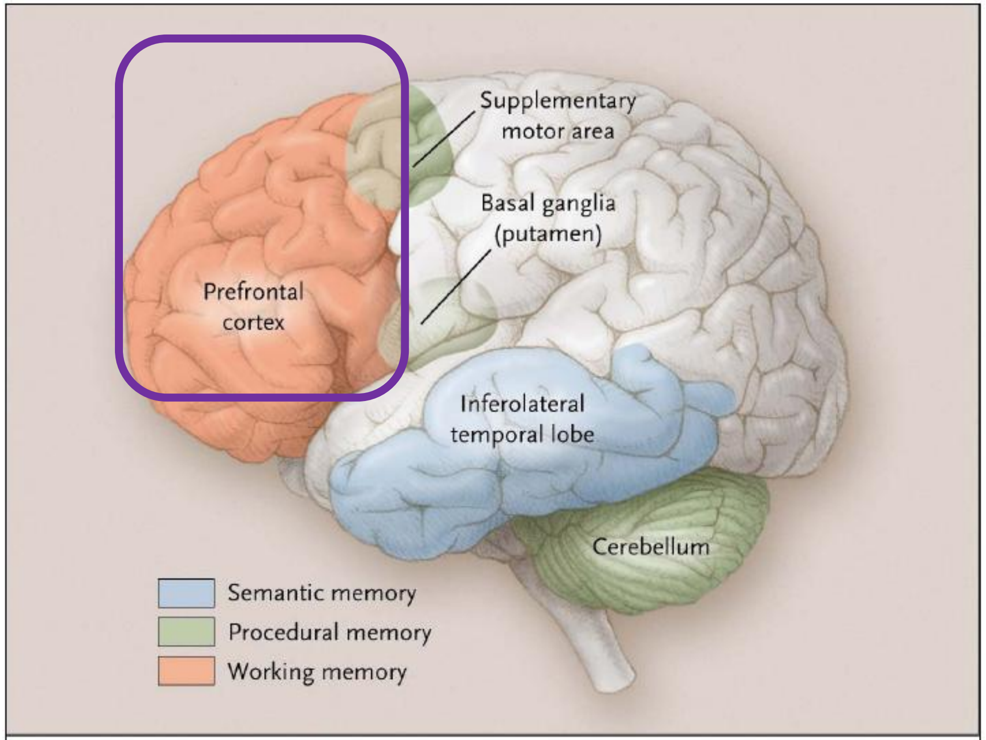


Figure 2. Semantic, procedural, and working memories. The inferolateral lobes are important in the naming and categorization tasks by which semantic memory is typically assessed. However, in the broadest sense, semantic memory may reside in multiple and diverse cortical areas that are related to various types of knowledge. The basal ganglia, cerebellum, and supplementary motor area critical for procedural memory. The prefrontal cortex is active in virtually all working memory tasks. Other cortical and subcortical brain regions will also be active, depending on the type and complexity of the working memory task. (Budson and Price, 2005; permission granted by the *New England Journal of Medicine*.)

The brain

The brain makes up 2% of a person's weight

Brain consumes 20% of the body's energy

- Even at rest

Brain energy consumption

- 10 times the rate of the rest of the body per gram of tissue

Average power consumption of a typical adult

- 100 Watts
- brain consumes 20% of this making the power of the brain 20 W

Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 104-105, 352ff

Working Memory (WM)

- memory storage for up to 30 seconds, after that short period, the contents decay or are displaced
- rapid access: ~70ms access time
- storage size: 3-4 “chunks” (NOT: 7+-2 elements!)*
- it is easy to overwrite the contents (intentionally and unintentionally!)
- can store two different types of data at the same time:
  - visual information (visuo-spatial sketchpad)
  - verbal information (articulatory loop)
- to maintain the contents of working memory: rehearsal is needed
- to persistently store information from working memory: move it to long-term memory

* Cowan, Nelson (2001). "The magical number 4 in short-term memory: A reconsideration of mental storage capacity". Behavioral and Brain Sciences **24** (1): 87–114; discussion 114–85
Designing Interactive Systems; D. Benyon, P. Turner and S. Turner; 2005, Pearson, p. 104-105, 352ff

Working Memory types

Verbal: Phonological Store

- Retrieval: Articulatory loop
- „Programmed“ and associated by words written, spoken (⍰ simpler to remember)

Visual

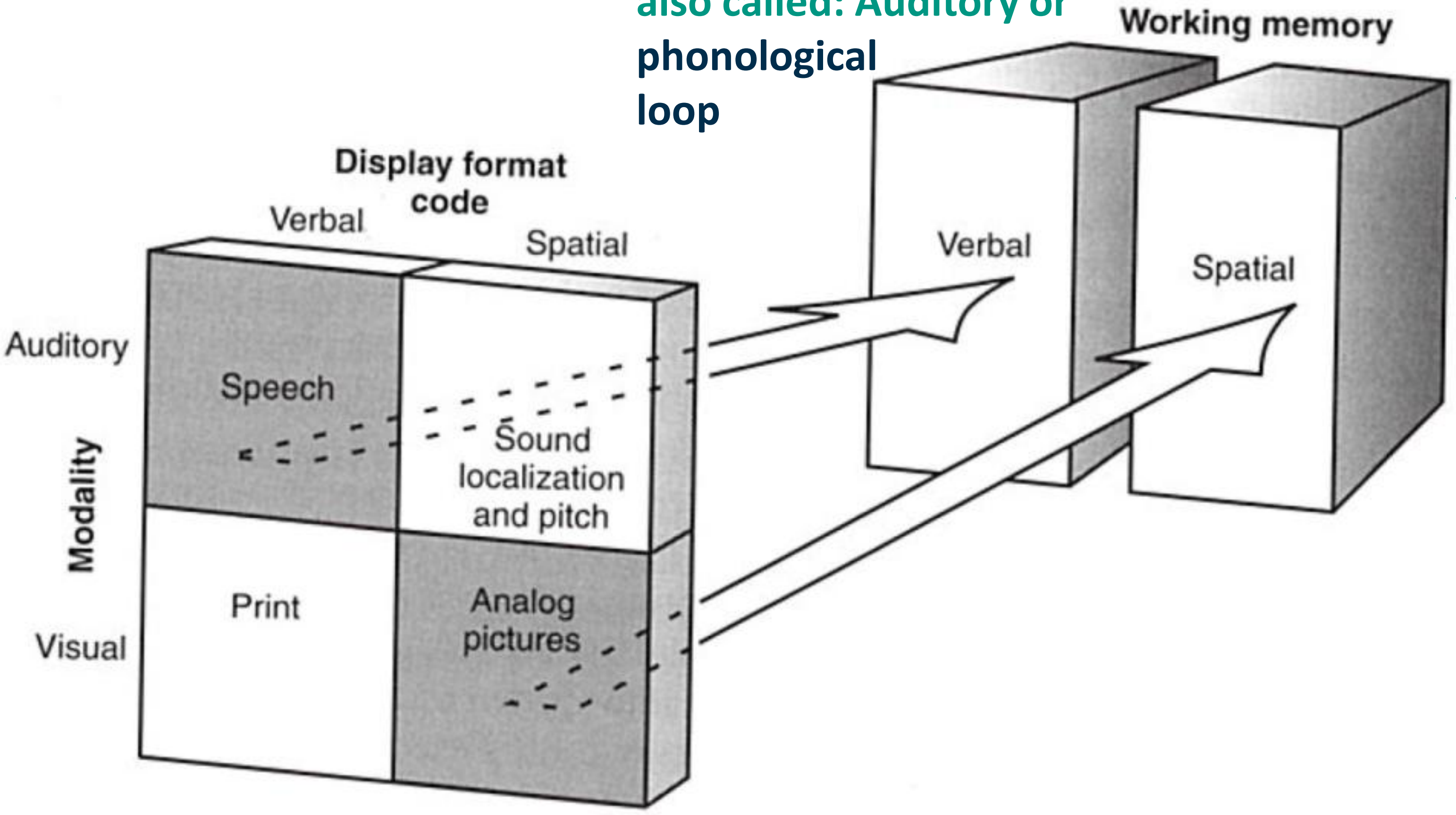
- Visuospatial Sketchpad
- Stored often in form of images

Operate in parallel

- But there is only a restricted number of attention levels

Code and Modality Mapping

also called: Auditory or phonological loop



Visuo-spatial Sketchpad

Long-Term Memory (LTM)

- “unlimited” in capacity
- lasts from a few minutes to life time
- slow access: 100ms seconds access time
- multi-modal memory
  - smell as strong trigger to long-term memory example
  - sound as most efficient cue for working memory
- three types (could be clustered otherwise, depending on psychological model)
  - episodic – serial memory of events
  - procedural – knowledge of how to do things
  - semantic – structured memory of facts, concepts, skills
- semantic LTM derived from episodic LTM

Memory Model and Aging

Memory

Working Memory

Long Term Memory

Declarative
(explicit):
Facts & events

Procedural
(implicit):
Motor Skills etc.
Stays the same with age
(but not your physical
condition)
American Physiological Organisation

Episodic
(incl. Autobiographic):
Experiences, events
and it‘s context
Decreases with age

Cognitive
Maps

Semantic
Facts, meaning,
concepts

Stays with age
(or improves)

Age invariance in semantic and episodic metamemory: Both younger and older adults provide accurate feeling-of-knowing for names of faces
Deborah K. Eakin, Christopher Hertzog, William Harris. Aging, Neuropsychology, and Cognition , 2013

Long-Term Memory (LTM): Inside

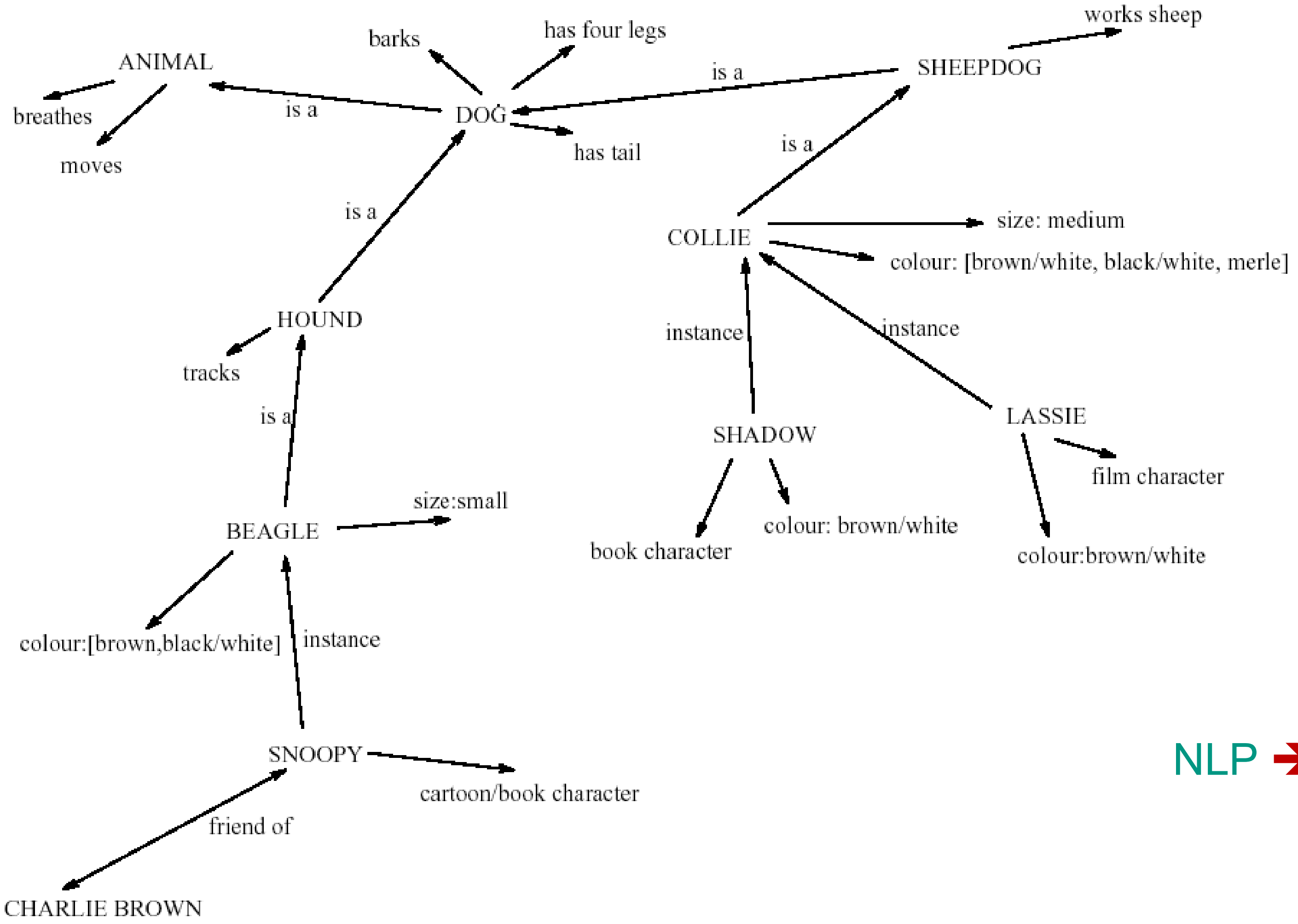
Semantic memory structure

- provides access to information
- represents relationships between bits of information
- supports inference

Model: semantic network

- inheritance – child nodes inherit properties of parent nodes
- relationships between bits of information explicit
- supports inference through inheritance

Long-Term Memory (LTM): Inside



NLP ➔ Embeddings

Long-Term Memory (LTM): Processes I

Forgetting:

- **decay:** information is lost gradually, but very slowly
- **Interference**
  - new information replaces old: retroactive interference
  - old may interfere with new: proactive inhibition
- affected by emotion – can subconsciously `choose' to forget

Long-Term Memory: Processes II

Encoding is the process by which information is stored in memory

Retrieval is the means by which memories are recovered from long-term storage

Forgetting is the name of a number of different possible processes by which we fail to recover information

Long-Term Memory: Processes III

recall ("Erinnerung"): active memory search to retrieve a particular piece of information

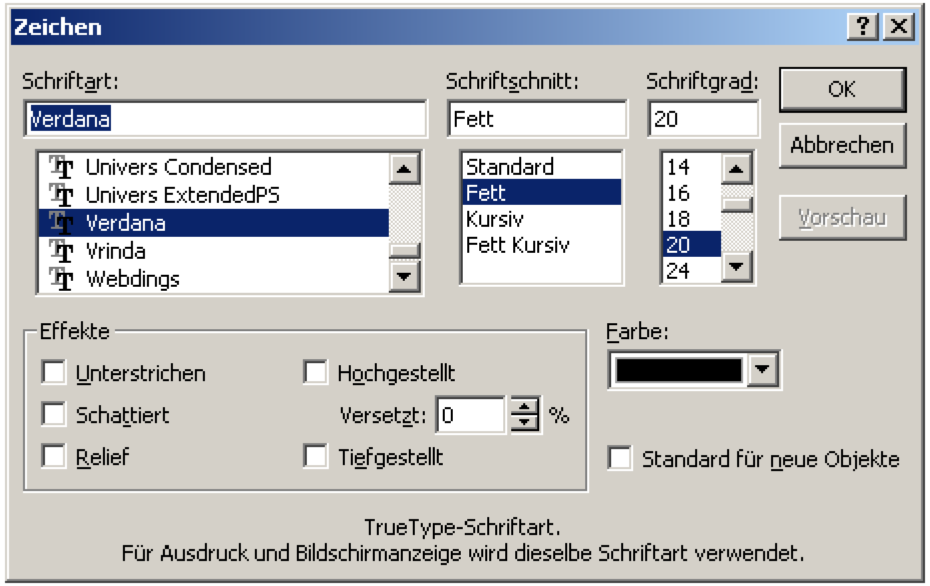
recognition ("Wiedererkennung"): searching the memory and deciding whether the retrieved piece of information matches a given information; recognition is generally easier and quicker than recall

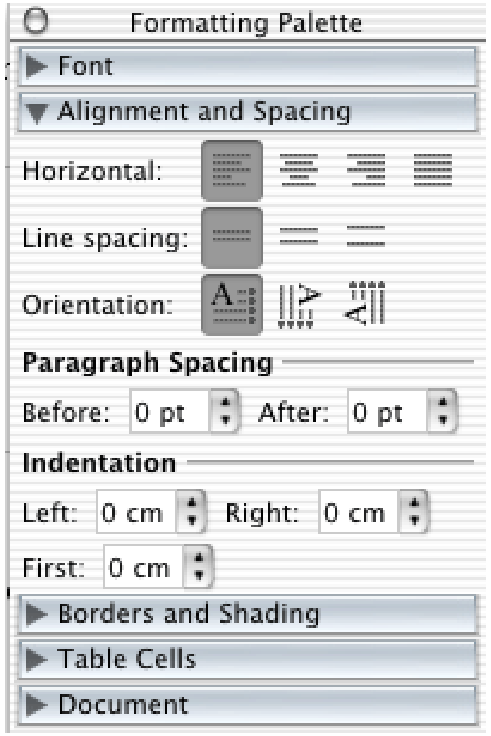
rehearsal:

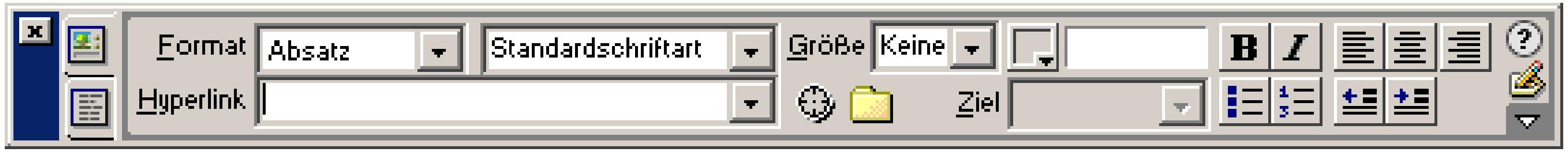
- **There is no direct link from perception to LTM!**
- The process of repeating information in working memory. This facilitates the short-term recall of information and its transfer to long-term memory
- the amount stored is proportional to the number of rehearsals!

chunking: grouping of items into more meaningful units

Issue: Recall, Recognition, Chunking







Prepare for The experiment Task: Remember the Number

Test – Memorize numbers

57324620

Test – Memorize numbers

Write down: What was the number?

Test – Memorize numbers

57324620

Prepare for The experiment Task: Remember the Number

Test – Memorize numbers

74 23 94 10

Test – Memorize numbers

Write down: What was the number?

Test – Memorize numbers

74 23 94 10

Time and recognition

Remembering information over a longer period of time is difficult

Recognition might be of help

- It is much easier to recognise something than to recall it
- This holds for images (this is the reason for thumbnails), but also for text
  - E.g. it is easier to recognise a (difficult) word than to remember
  - Stanstead, Stansted or Standsted?
  - Simple with a selection list of all London airports.
- Web-Pages are more difficult to recall, because they change format. Recall is often also bound to the shape.

Icons

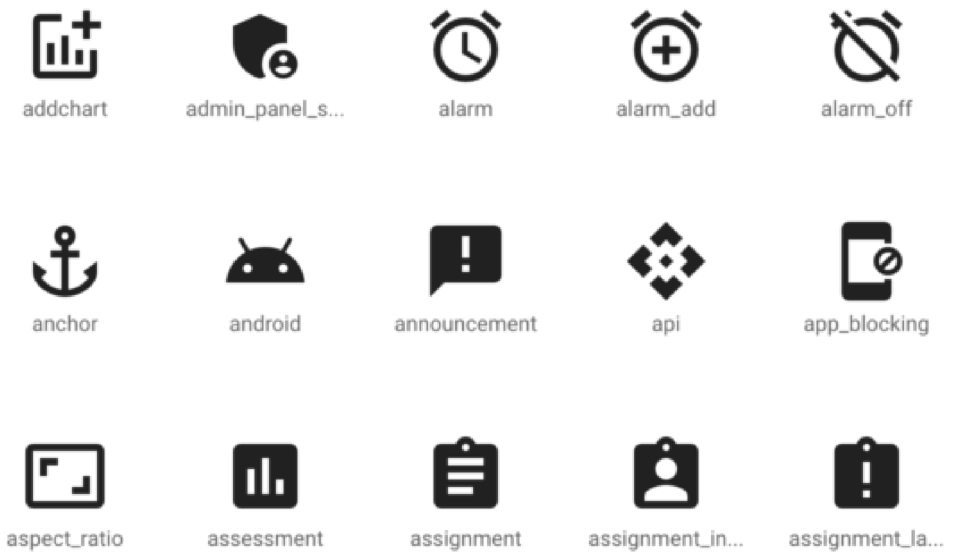
Icons often use metaphor to recall an associated object or activity

There are only a few directly associable icons

– Others are introduced by someone, often large companies

There is Horton‘s checklists for icons:

– Understandable, Familiar, Unambiguous, Memorable, Informative (concept important?), few (<20), distinct, attractive, legible (tested in all view contexts), compact (means minimal) coherent (can be separated from other icons and the background), extensible (means size scalable)



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Memory Design: Guidelines

M1: Organize information into a small number of "chunks"
M2: Try to create short linear sequences of tasks
M3: Use persistence, so do not flash important information onto the screen for brief time periods
M4: Do not "overwrite" the contents of working memory by giving additional tasks to the user
M5: Organize data fields to match user expectations or to organize user input (e.g. the automatic formatting of phone numbers)
M6: Provide reminders or warnings of the state the user has reached in an operation
M7: Provide ongoing feedback on what is happening and/or what has just happened
M8: The user interface should behave in consistent ways at all the times for all screens
M9: Terminology, icons and the use of colour should be consistent between screens

Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 390

Guidelines, Principles and Theories for HCI

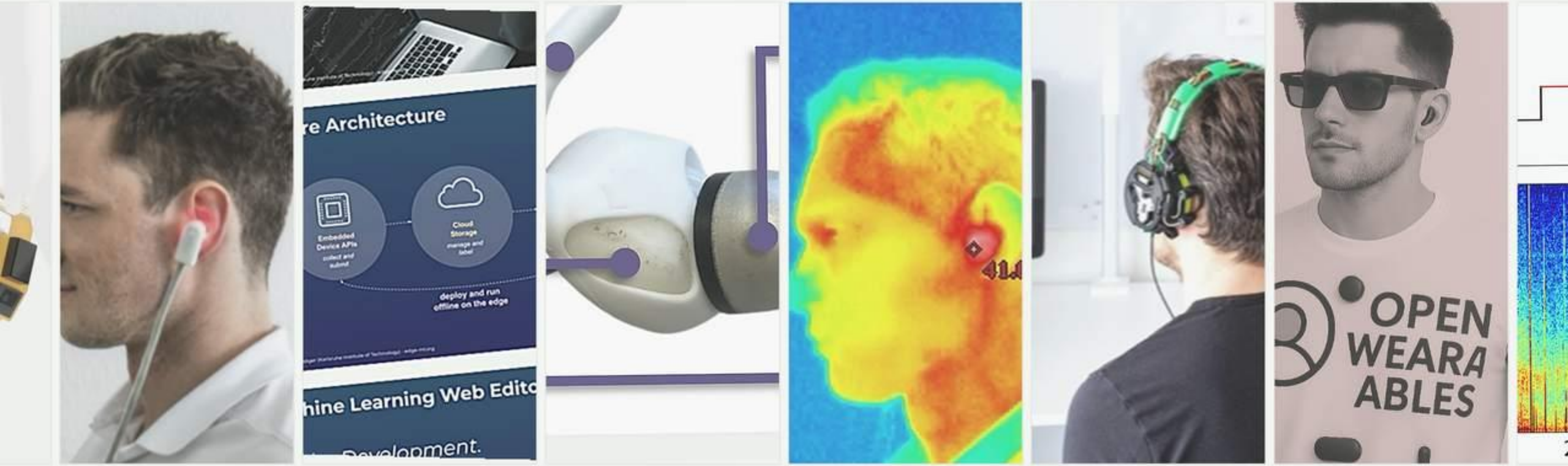
Guideline: specific and practical rules for solving problems of UI design

Principles: help for analyzing and comparing design alternatives

Theories and Models: describe objects and actions with consistent terminology and give a comprehensible explanation of connections

Pingo

Question:
Answer question about “Recognition/Recall”
Link:
https://pingo.coactum.de/589809



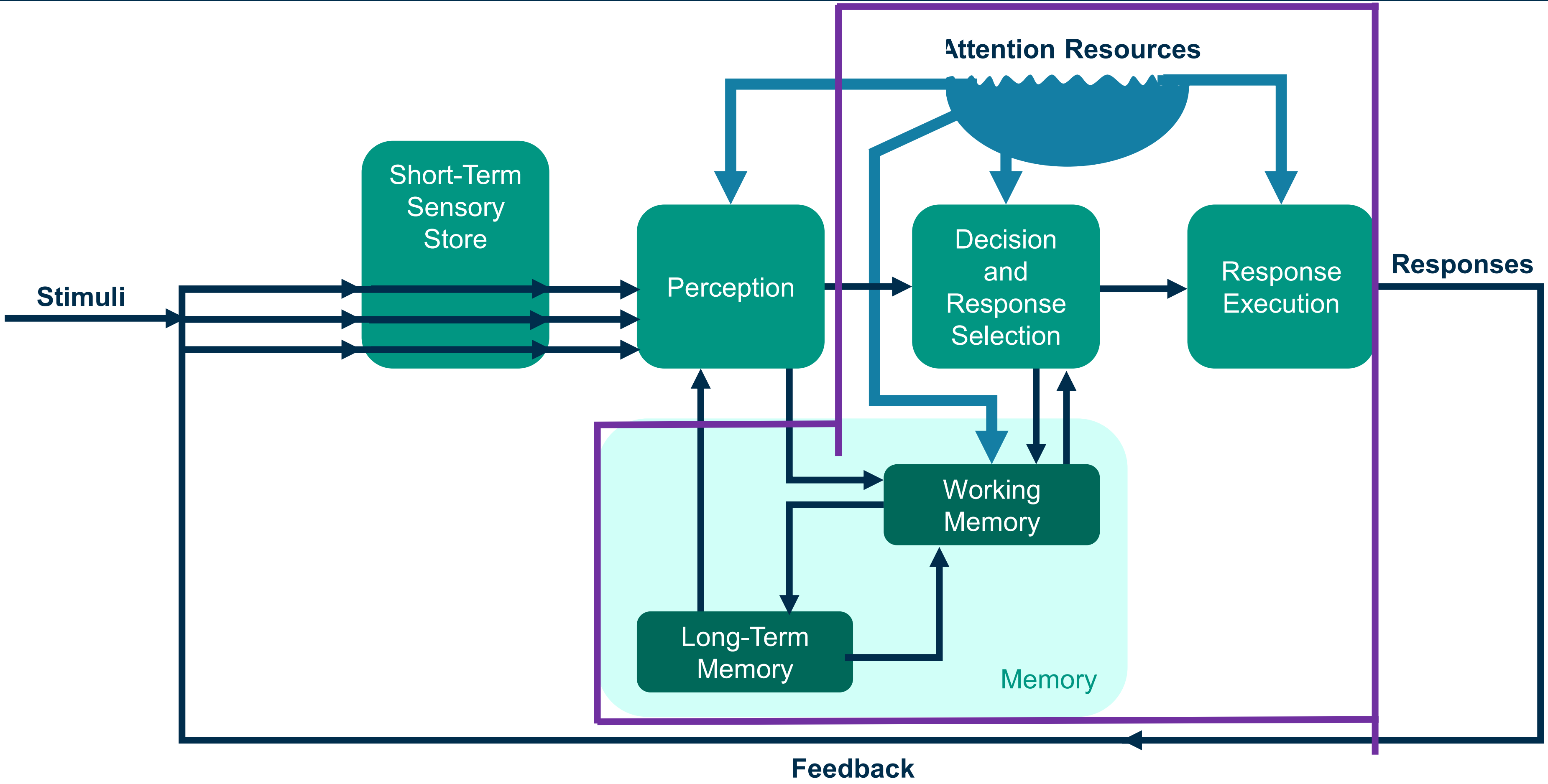
Human Computer Interaction SS26

Chapter
2
HIP

Michael Beigl, TECO

Decision →

Decision Making



Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Prepare for The experiment
Task: Remember the time
used

→

Test – Motivation for Decision Making and Long-Term Memory

Do dogs bark?

Bellen Hunde?

Test – Motivation for Decision Making and Long-Term Memory

Do dogs breathe?

Atmen Hunde?

- The second question takes longer to answer
- this indicates semantic coding!
- without memory, there is no “thinking” or decision-making!

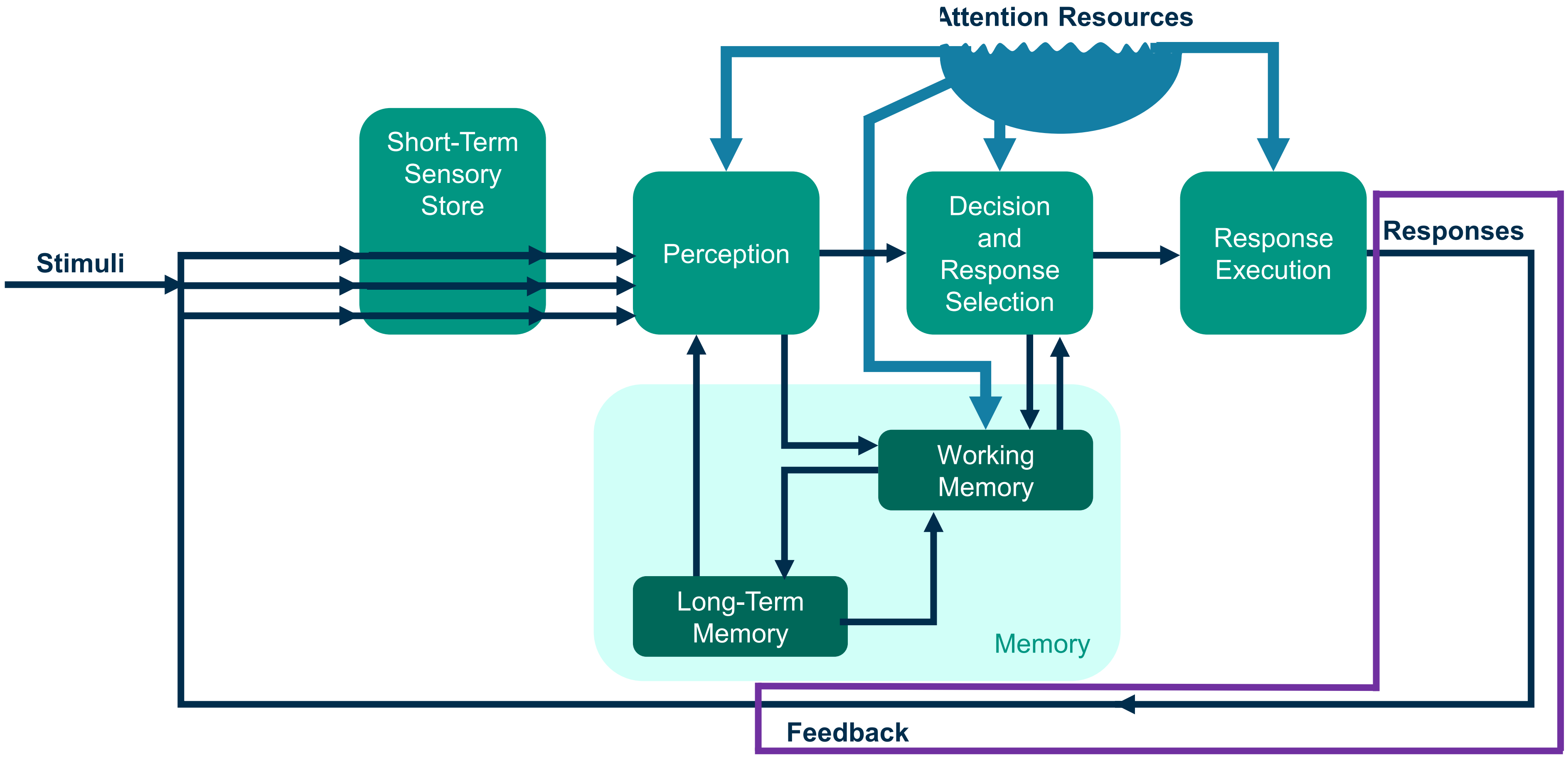
Decision Making

after the perception of the stimulus, a response needs to be selected

Automatic vs. controlled decisions

- automatic: fast
  - little or no attention required
  - learned reflexes or behavior
  - a long-term memory procedure is executed nearly automatically in response to the stimulus
- controlled: slow
  - attention required, typically conscious of thoughts
  - interaction with working memory (WM) and long-term memory (LTM) systems
  - cf. “planning problem” e.g. from robotics (top-down or bottom-up, task separation)

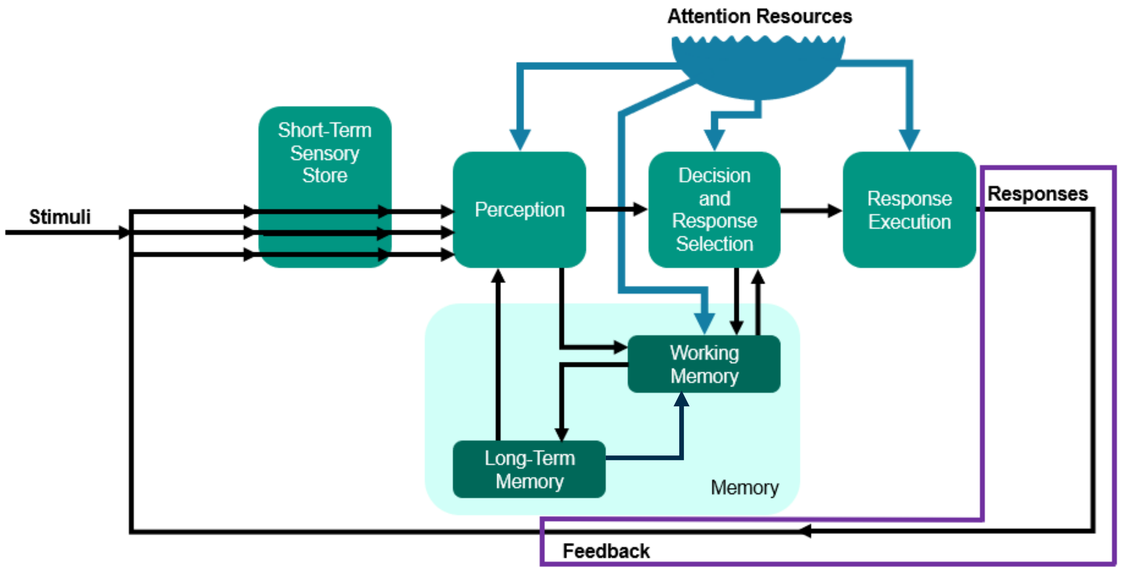
Response and Feedback



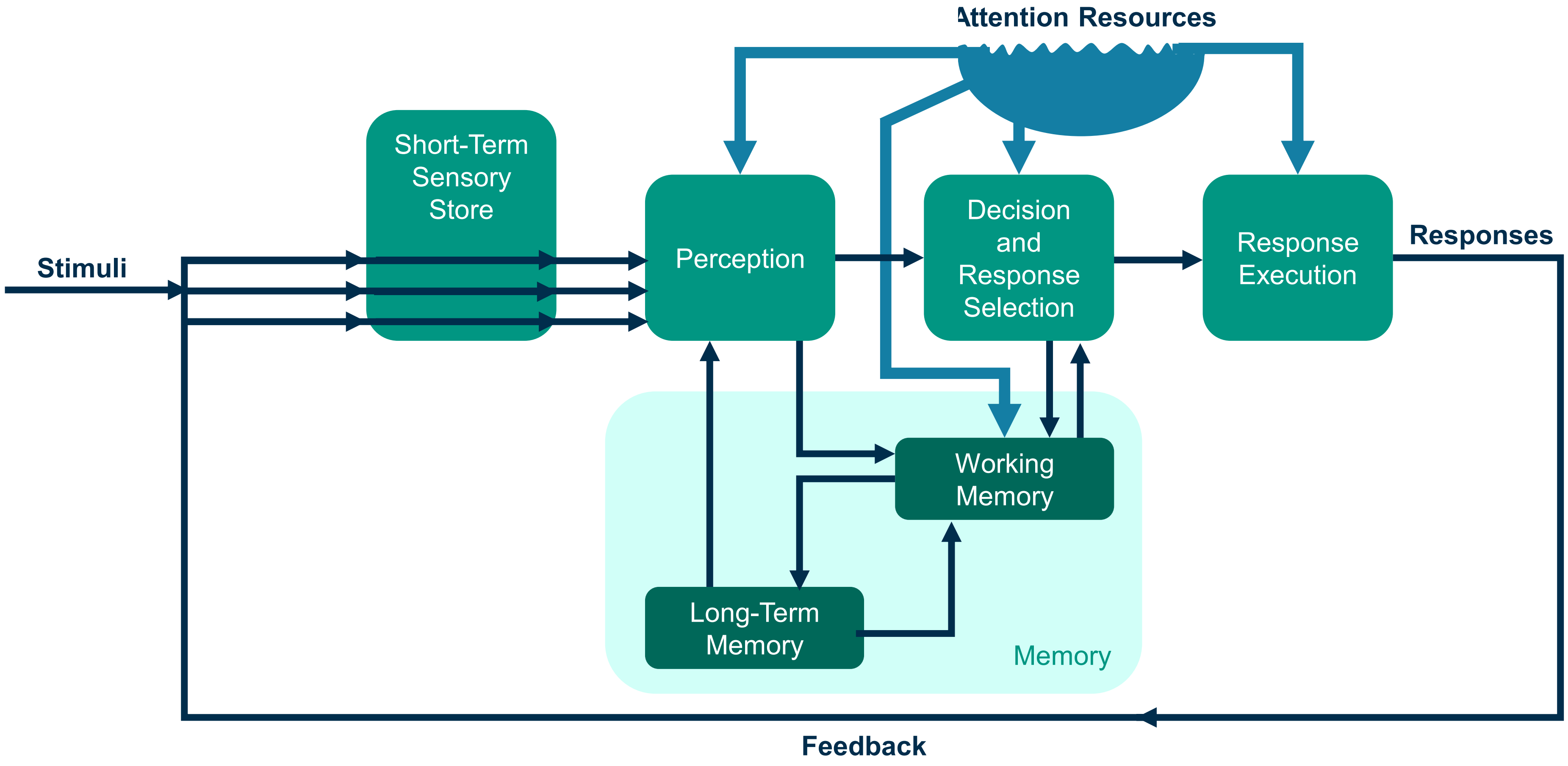
Engineering Psychology and Human Performance; C. Wickens and J. Hollands, 1999, Prentice Hall, p. 11, graphically changed

Response and Feedback

- after a decision has been made, it has to be executed by complex motor movements
- feedback-loop: we observe the consequences of our actions, producing closed-loop feedback
- the model is circular rather than linear (see HIP model)



Human Information Processing



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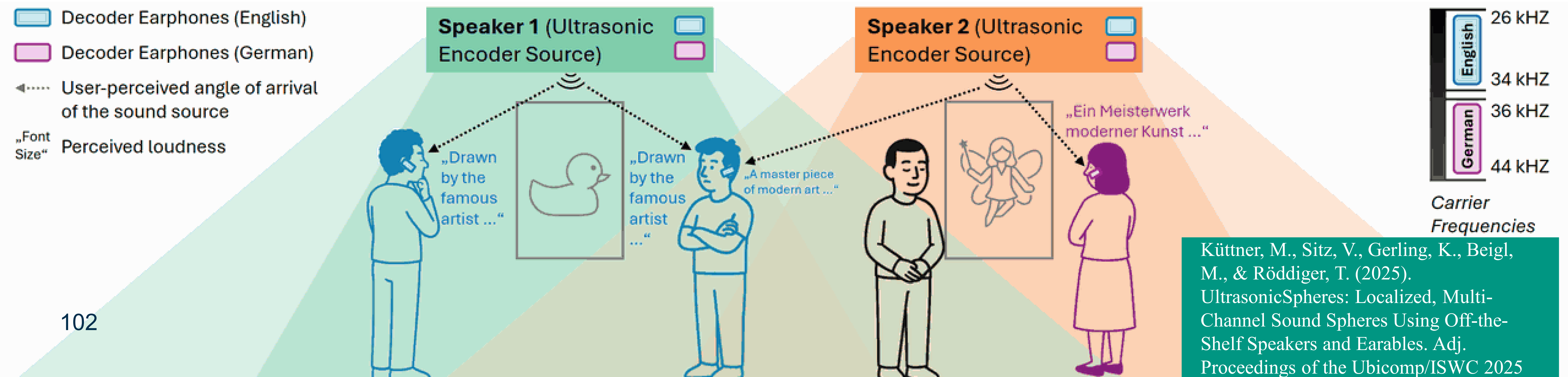
UltraSpheres →

Küttner, M., Zitz, V., Gerling, K., Beigl, M., & Röddiger, T. (2025). UltrasonicSpheres: Localized, Multi-Channel Sound Spheres Using Off-the-Shelf Speakers and Earables. Adj. Proceedings of the Ubicomp/ISWC 2025

Can we generate natural, but personalized spatial audio experience?

Ultrasonic Spheres are localized, directed, “distance aware” ultrasonic audio zones decoded by earphones, providing personal, spatially targeted sound



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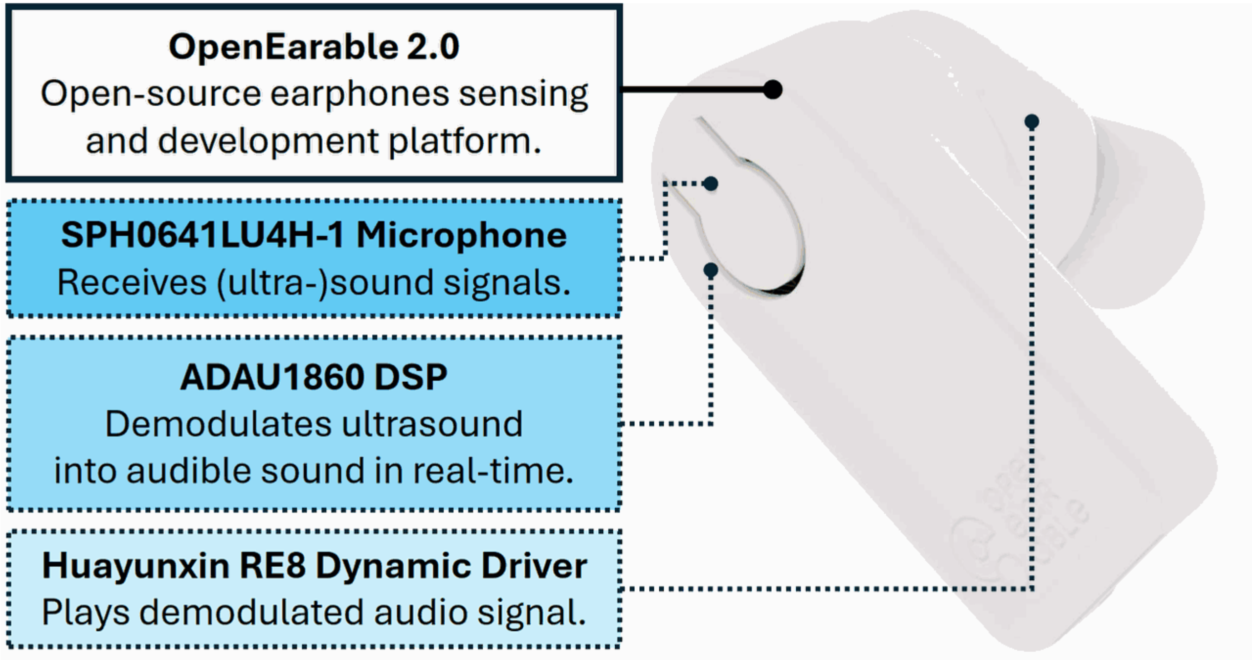
What is needed for a setup?

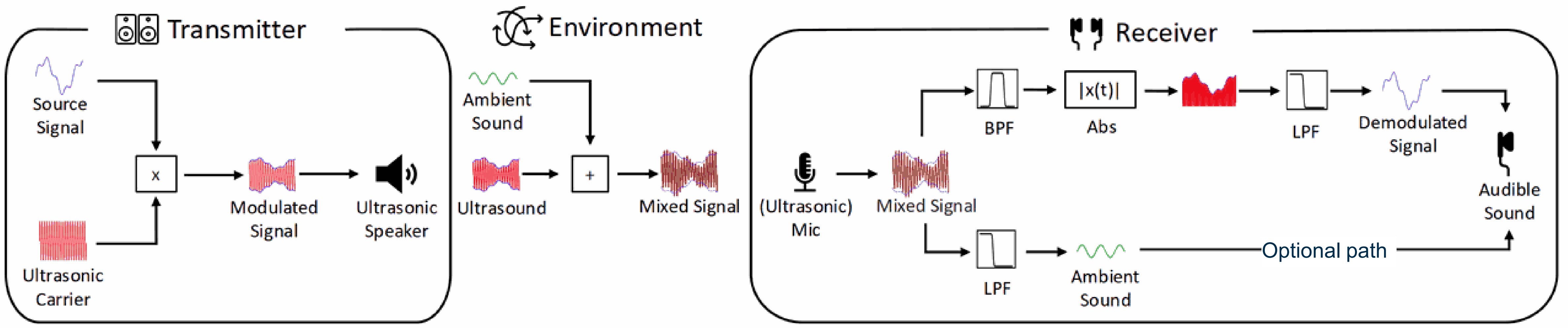
- A Earable device capable of receiving US and processing signals
- A loudspeaker capable of emitting Ultrasonic waves

Küttner, M., Sitz, V., Gerling, K., Beigl, M., & Röddiger, T. (2025). UltrasonicSpheres: Localized, Multi-Channel Sound Spheres Using Off-the-Shelf Speakers and Earables. Adj. Proceedings of the Ubicomp/ISWC 2025



How does it work technically?

- At the speaker, audio is mixed with an US carrier
- In the environment, audio and ambient sounds mix
- At the receiver, ambient and US signals are separated
- The US mixed audio (source audio) is filtered and demodulated



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Why is the magic?

On the human side two “audio processors” work in parallel

- The motor processor to direct your ear to the wanted sound source
- The cognitive processor, separating various incoming audio streams, focusing attention to one or the other

The behavior is natural and mostly sub-conscious

Küttner, M., Sitz, V., Gerling, K., Beigl, M., & Röddiger, T. (2025). UltrasonicSpheres: Localized, Multi-Channel Sound Spheres Using Off-the-Shelf Speakers and Earables. Adj. Proceedings of the Ubicomp/ISWC 2025

CMN Model:

HIP by Numbers
The Model Human Processor (MHP)

CMN-Model Inventors (1983)

Stuart Card

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Thomas Moran

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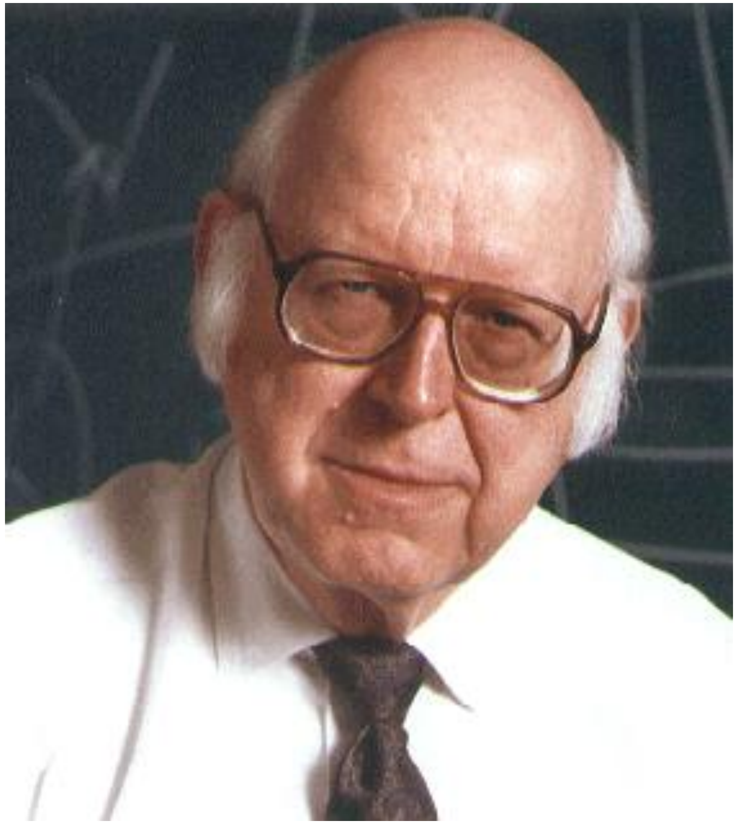
Allen Newell



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CMN-Model

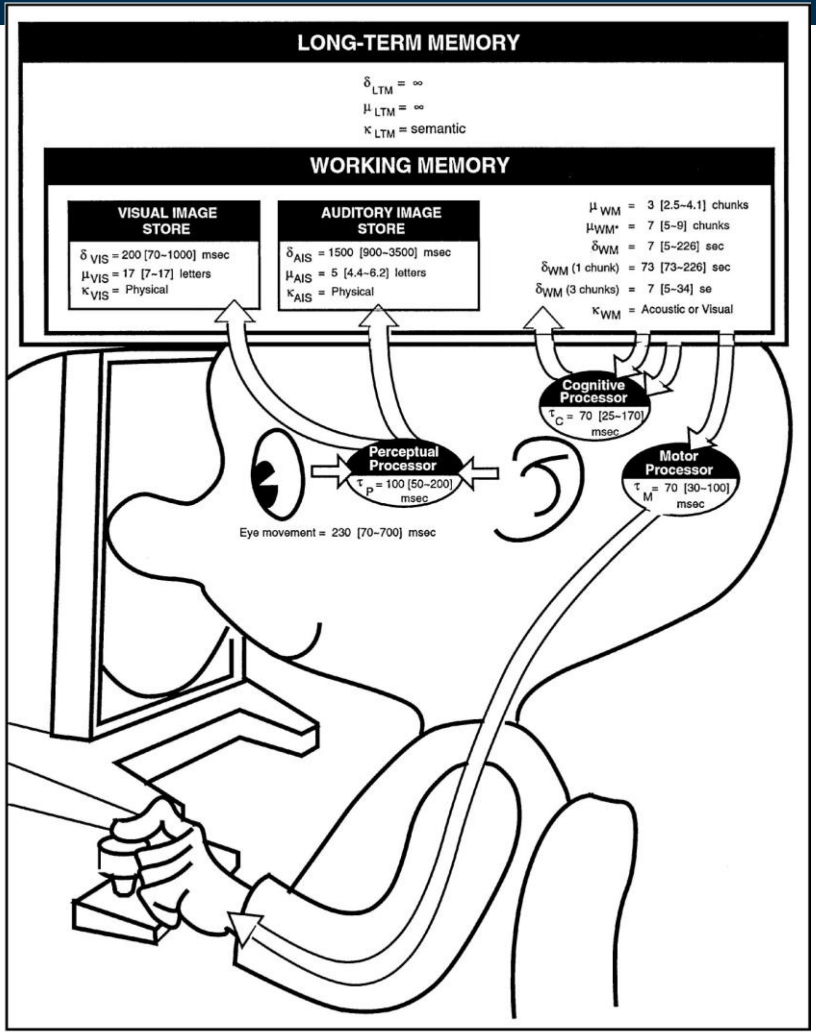
most influential model of user interaction

3 interacting subsystems

- Perceptual, cognitive & motor
- each with processor & memory
  - described by parameters (e.g. capacity, cycle time)
- serial & parallel processing

principles of operation

- subsystem behaviour under certain conditions
  - e.g. Fitts’s Law, Power Law of Practice



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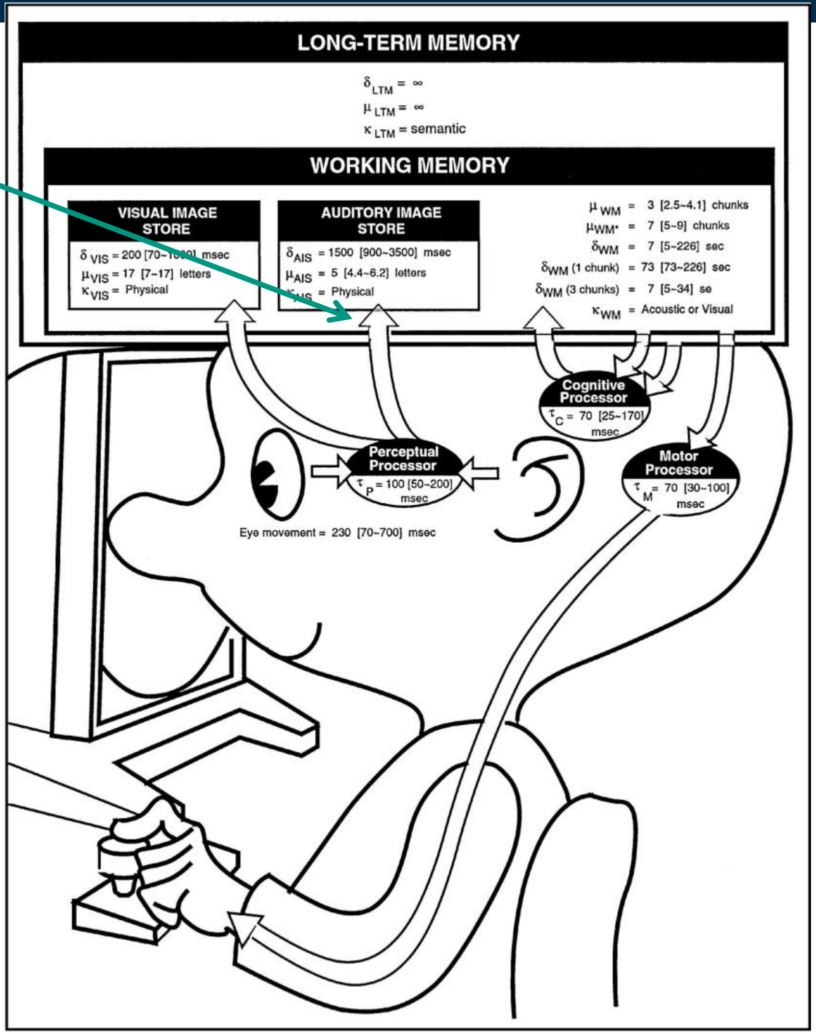
Perceptual Processor



Perceptual processor

- sensory input (audio & visual)
- code info symbolically
- output into audio & visual image storage (WM buffers: visual sketchpad, auditory loop)

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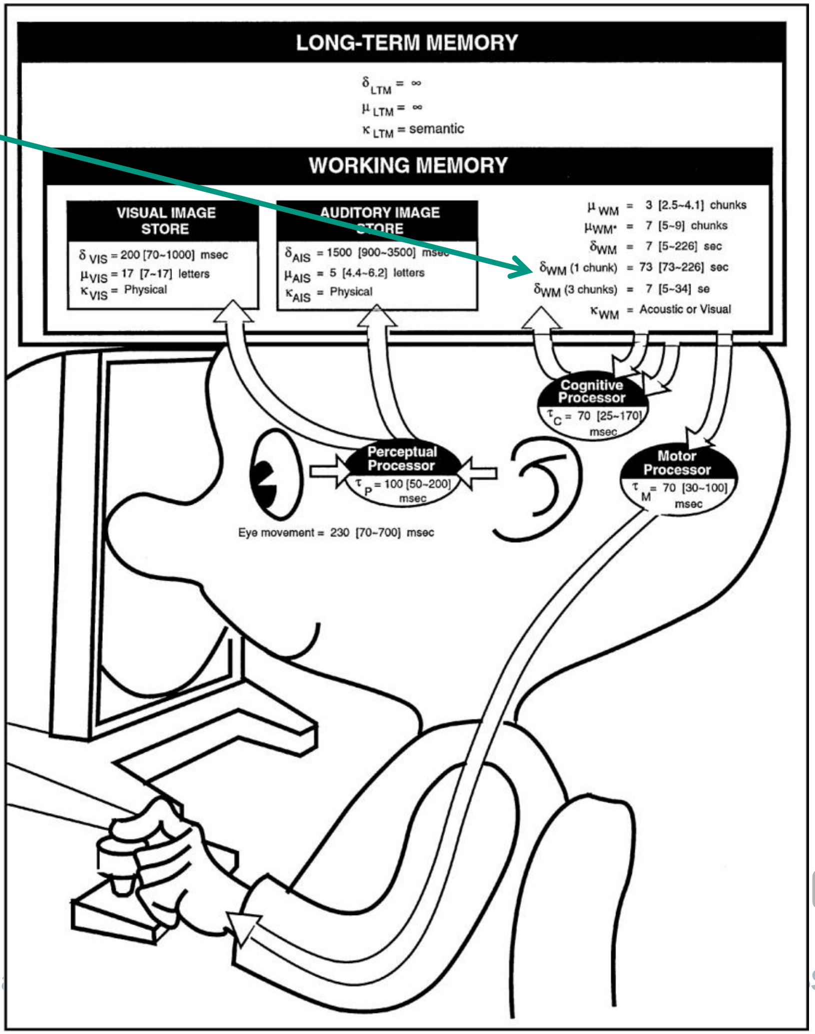
Cognitive Processor

Cognitive processor

- input from sensory buffers
- access LTM to determine response
  - previously stored info
- output response into WM



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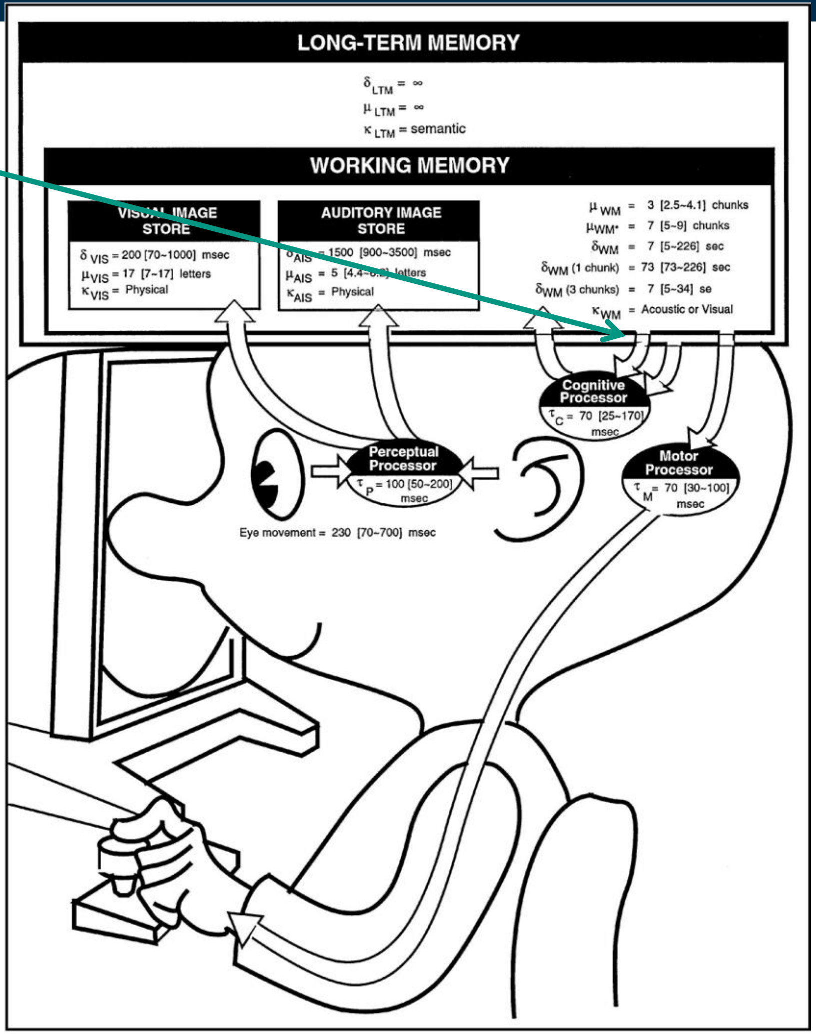
Motor Processor

Motor processor

- input response from WM
- carry out response



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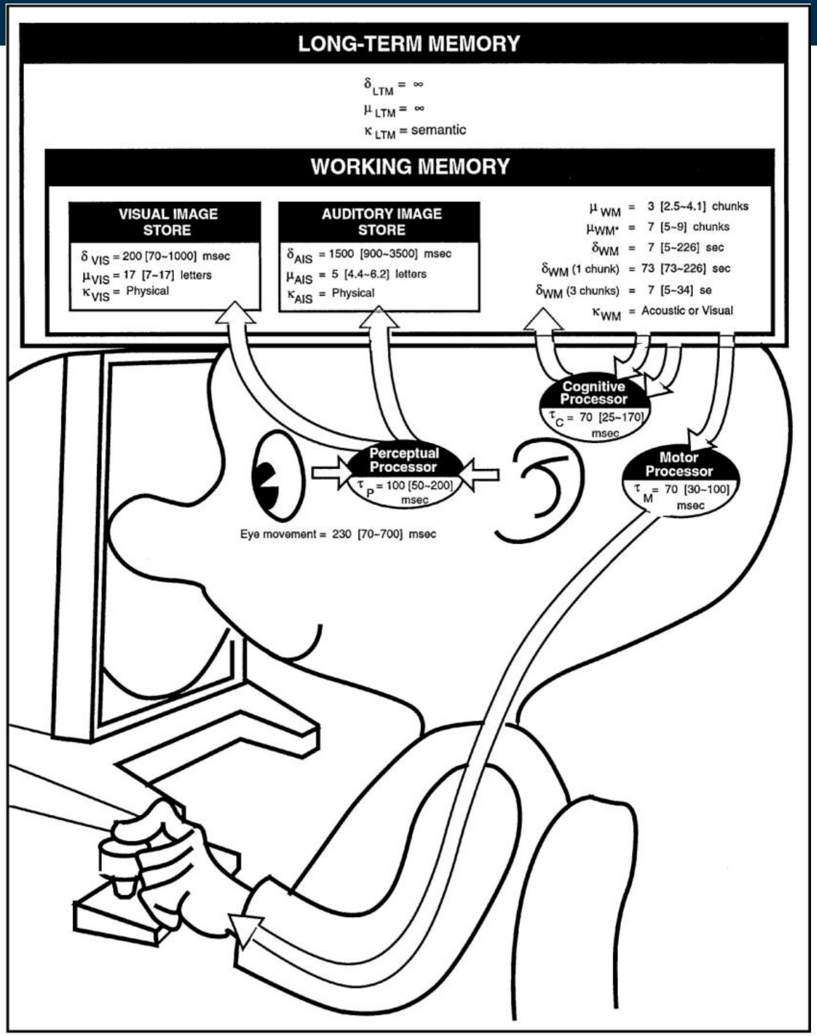
Interactions

Input/output

Processing

- serial action
  - pressing key in response to light
- parallel perception
  - driving, reading signs & hearing



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Parameters

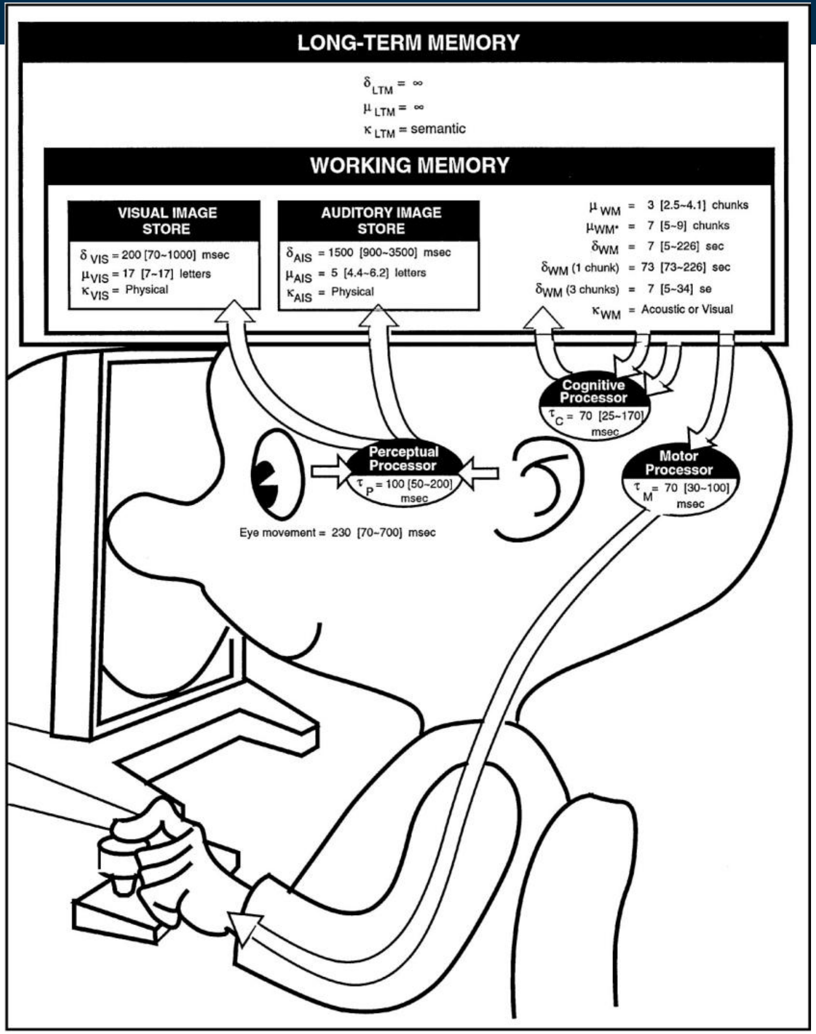
Based on empirical data

- word processing in the ‘70s

Processors have **cycle time (τ)**

Memories have

- **storage capacity (μ)**
- **decay time** of an item **(δ)**
- **info code type** **(κ)**
  a. physical, acoustic, visual & semantic



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Parameters

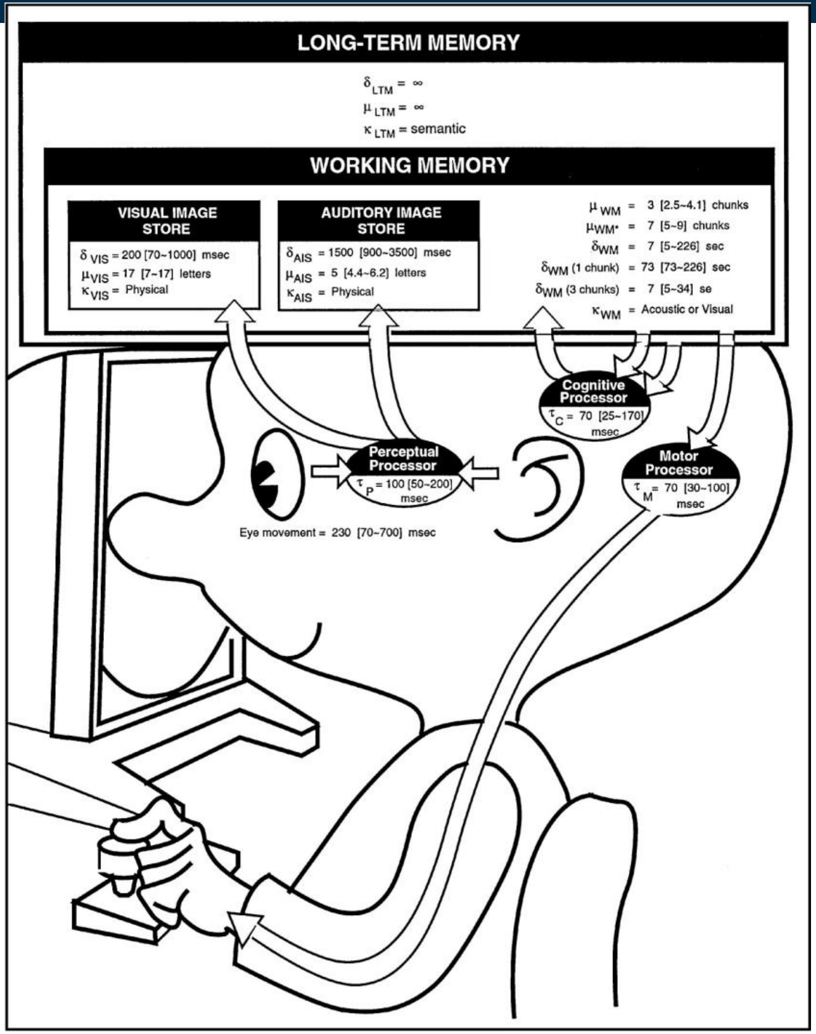
Not fixed numbers, all “about”

Processor Cycle time (τ)

- perceptual = 100 msec, cognition = 70 msec and motor processing = 70 msec
- Total time: τ(per)+ τ(cog)+ τ(mot)

Visual Image Store

- Storage capacity (μ) perception
  - = 17 letters visual
  - = 5 letters audio
- storage capacity (μ) working memory
  - = 3 chunks (number of digits recalled from an unexpected interrupt string)
  - = 7 chunks (working memory in conjunction with LTM)



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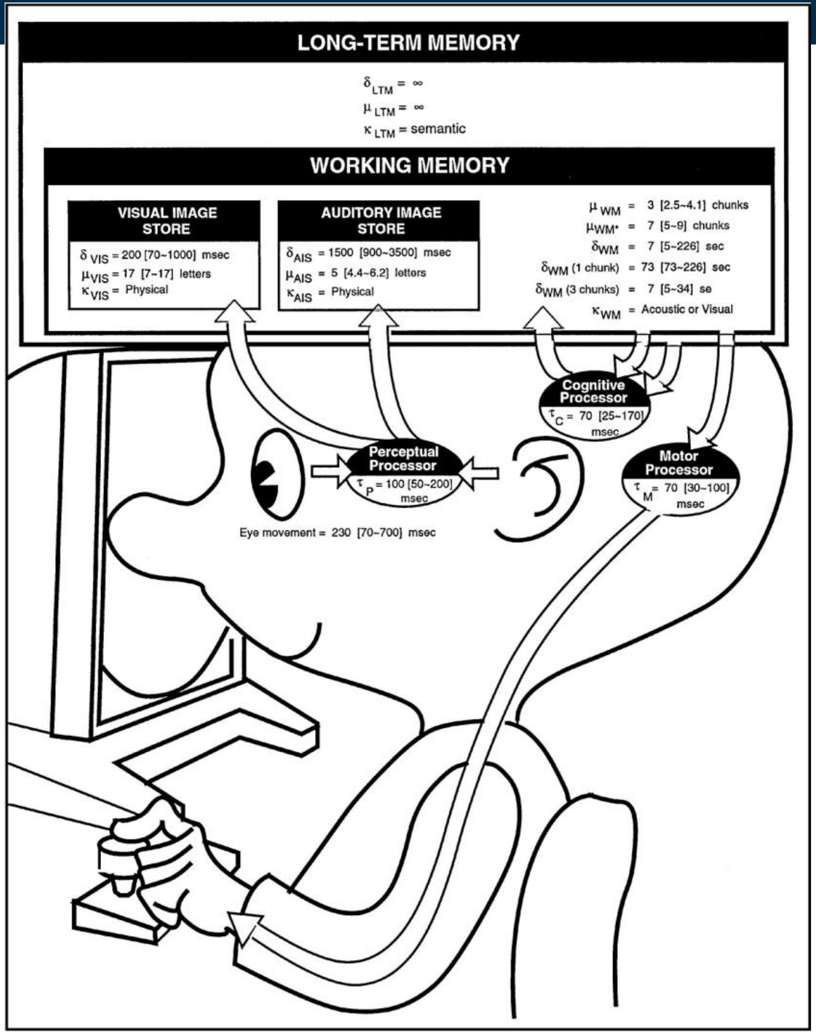
Parameters

- Decay time of an item perception (δ)
  - = 200 msec visual (half-life index)
  - = 1500 msec audio
- Decay time of Working Memory (δ)
  - = 7 Sec for 3 chunks
  - = 73 (ca. 70) seconds for 1 chunk

- info code type (κ) = physical
  - physical properties of visual stimulus
    - e.g. intensity, color, curvature, length



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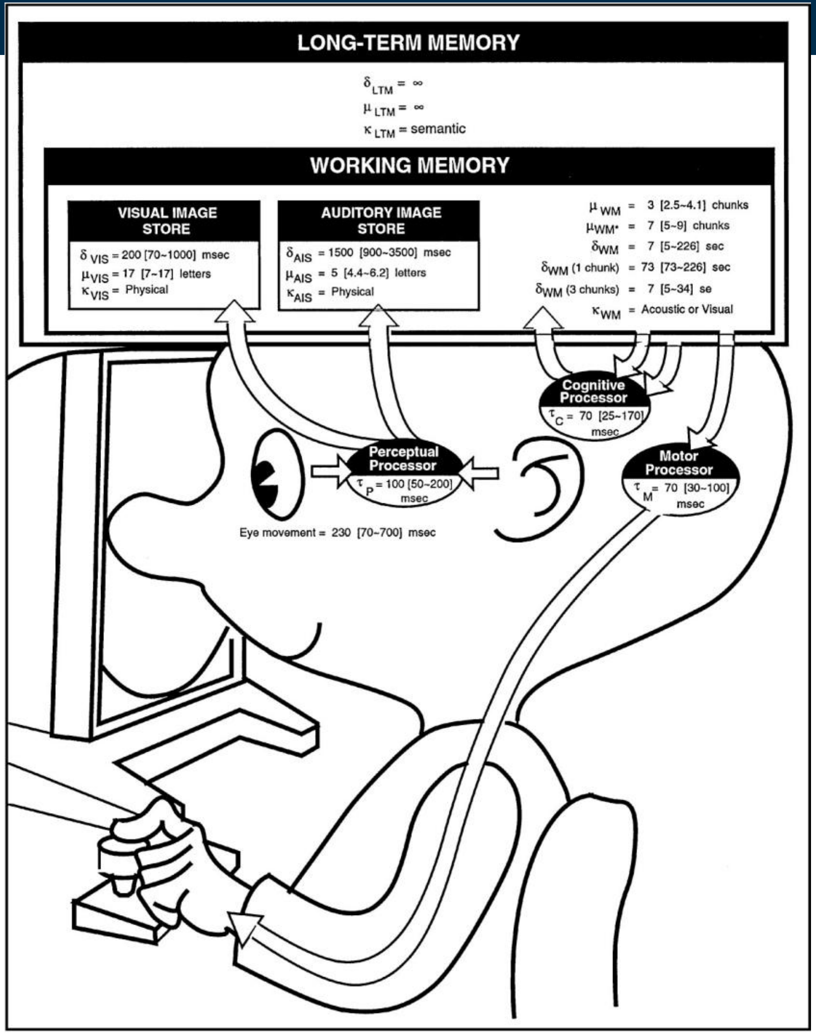
One Principle of Operation

Power Law of Practice

- task time on the nth trial follows a power law
  - $T_n = T_1 n^{-a}$, where a = .4
  - i.e., you get faster the more times you do it!
  - applies to skilled behavior (perceptual & motor)
  - does not apply to knowledge acquisition or quality



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HIP by numbers

What is the **use for these numbers**?

- These numbers are used later for the Interaction design models
- There will be more numbers and formula, that let us describe specific human performance activities

Pingo

Question:
Answer question about “CMN”
Link:
https://pingo.coactum.de/589809